Computational fluid dynamics, multi-species canopy gas exchange, and harvest fixation architecture of the LEAF GBC1 lunar plant growth chamber

Richard Barker[1*]

[1*]Department of Agricultural and Biological Engineering, Purdue University, 225 S. University Street, West Lafayette, 47907, IN, USA.

Corresponding author(s). E-mail(s): barkerrj@purdue.edu;

Abstract

Plant cultivation on the lunar surface during the Artemis III mission demands precise microenvironmental control within sealed growth chambers to counteract the loss of natural buoyant convection. Here, we present a high-fidelity computational fluid dynamics (CFD) and biological transport model of the Lunar Effects on Agricultural Flora (LEAF) Growth Bed Chamber 1 (GBC1; CAD assemblies M09024F-MDL-5 and M09025F-MDL-2). We resolve Navier–Stokes and Brinkman–Darcy flow across 164 individual plant crowns (*Brassica rapa* and *Arabidopsis thaliana*) co-cultivated within a porous polyurethane and Nytex substrate. Under lunar gravity ($\boldsymbol{g_{\mathrm{lunar}} = 1.62\ \mathrm{m\,s^{-2}}}$), buoyant thermal convection decreases by 83.4% relative to terrestrial 1g controls, expanding the foliar boundary layer and throttling transpiration and carbon assimilation. Leaf boundary-layer conductance ($\boldsymbol{g_{\mathrm{b}}}$) is estimated from closed-form correlations rather than resolved by the flow solver, and each solver run covers only **1.8%** of one headspace turnover, so the model cannot yet size a compensatory pump flow: across $\boldsymbol{Q = 0.05}$–$\boldsymbol{4.0\ \mathrm{L\,min^{-1}}}$ the lunar correlation estimate ($\boldsymbol{\leq 0.054\ \mathrm{mol\,m^{-2}\,s^{-1}}}$) never reaches the Earth value ($\boldsymbol{\geq 0.0715}$). We model nocturnal respiratory $\mathbf{CO_2}$ accumulation, diurnal day/night photoperiod cycling, optical lid condensation prevention ($\boldsymbol{\Delta T_{\mathrm{margin}} \geq +0.83^{\circ}\mathrm{C}}$), and terminal harvest fixation via bottom-up RNAlater perfusion (**31.7 s** root immersion). A three-grid study does not establish grid independence, because the grids are compared at different simulated times. These results are a model-development stage, not flight rules for the LEAF GBC1 payload.

Keywords: Microgravity plant biology, Computational fluid dynamics, Lunar plant chamber, LEAF GBC1, Artemis III, Boundary layer conductance, RNAlater perfusion

1 Introduction

The exploration of the lunar South Pole under the NASA Artemis program marks a transition from low Earth orbit (LEO) orbital stations to sustained planetary surface exploration. As humanity prepares for extended crewed missions, biological life support systems, food supplementation, and fundamental plant physiological research must operate under hypogravity ($g_{\mathrm{lunar}} = 0.166g = 1.62\ \mathrm{m\,s^{-2}}$) and deep-space radiation regimes [1, 2]. The Lunar Effects on Agricultural Flora (LEAF) payload is an Artemis III deployed instrument that will grow *Brassica rapa*, *Wolffia* and *Arabidopsis thaliana* in a plant growth chamber with a crewed-cabin-like atmosphere, to study

germination, growth, gas exchange and genome-wide molecular responses to the lunar environment [1]. This study models one of its growth chambers, Growth Bed Chamber 1 (GBC1).

A fundamental biophysical challenge of plant growth in reduced gravity is the drastic alteration of fluid mechanics in the canopy and root zone [3, 4]. In terrestrial 1g environments, metabolic heat dissipation and foliar transpiration generate localized density gradients in the ambient atmosphere. These gradients drive free natural convection, continuously sweeping away warm, humid air and resupplying carbon dioxide (CO_2) to the photosynthetically active leaf surfaces [4–6]. In hypogravity, the Grashof number (Gr), which characterizes the ratio of buoyant forces to viscous forces:

$$Gr = \frac{g\beta\Delta T L^3}{\nu^2} \tag{1}$$

(where g is gravitational acceleration, β is the thermal expansion coefficient of air, ΔT is the surface-to-air temperature differential, L is the leaf characteristic dimension, and ν is kinematic viscosity) decreases by 83.4% on the Moon. Consequently, natural buoyant convection is largely extinguished. Without adequate forced ventilation, plants can become enveloped in thick, stagnant boundary layers, starving the chloroplasts of substrate CO_2, suppressing transpirational cooling, and inducing foliar heat stress, stomatal closure, hypoxia, and aberrant developmental phenotypes [3, 4, 7].

Previous spaceflight cultivation hardware—including the Biological Research in Canisters (BRIC) [8], the Advanced Plant Experiment (APEX) series, the European Modular Cultivation System (EMCS), the Advanced Biological Research System (ABRS), and the Veggie hardware aboard the International Space Station (ISS) [9, 10]—have addressed boundary layer stagnation with varying degrees of forced air movement. However, the LEAF GBC1 chamber presents unique design constraints:

1. Strict payload mass, volume, and power envelopes dictated by lunar lander egress and extravehicular activity (EVA) deployment.
2. Co-cultivation of multiple plant species displaying distinct canopy heights, leaf area indices (LAI), and aerodynamic roughness lengths: 32 robust *Brassica rapa* crop seedlings and 100 *Arabidopsis thaliana* genetic model seedlings situated across three discrete compartments separated by solid internal partition baffles (Chamber 1: 32 *B. rapa*, Chamber 2: 100 *Arabidopsis*, Chamber 3: 32 *B. rapa*).
3. A sealed, multi-chamber fluidic architecture requiring active water activation of dry polyurethane foam, five days of regulated vegetative growth, and autonomous terminal chemical fixation with RNAlater [11, 12] prior to lunar retrieval.

To ensure mission success, flight operations require rigorous computational modeling to establish the aerodynamic operating envelope, quantify ventilation uniformity, prevent optical lid condensation that would blind down-looking payload cameras, and optimize the perfusion kinetics of the fixative solution. In this study, we present a full 3D and 4D computational fluid dynamics (CFD), multi-phase porous media, and biological transport model of the LEAF GBC1 flight chamber (pot body model `M09024F-MDL-5` and transparent lid model `M09025F-MDL-2`). We resolve the Navier–Stokes equations coupled with Brinkman–Darcy flow through the porous growth matrix, Richards capillary imbibition, Farquhar–von Caemmerer–Berry (FvCB) photosynthetic kinetics, and closed-form estimates of leaf boundary-layer conductance (Section 4.4). The boundary-layer conductance is a correlation estimate, not a CFD result, and the flow solver is not run to steady state (Section 4.6); this version of the model therefore cannot establish the pump flow that would compensate for the 83.4% loss of buoyant driving force on the Moon.

—

2 Results

2.1 CAD-Faithful Chamber Geometry and Aerodynamic Architecture

The LEAF GBC1 growth vessel was reconstructed from exact engineering CAD models (`M09024F-MDL-5` for the lower fluidic pot body and `M09025F-MDL-2` for the transparent optical lid assembly). The chamber has a CAD headspace air volume of 54.05 cm^3 (repository README),

bounded by an aluminum housing, a lower fluid distribution plenum, a 15 mm thick open-cell polyurethane foam growth matrix topped with a Nytex mesh seed retainer, and an upper headspace enclosed by a high-transmittance polycarbonate optical lid (Fig. 1).

The canopy layout accommodates 164 individual seeds arranged in two distinct physiological zones:

- **Zone A (*Brassica rapa*)**: 32 seeds arranged in a 4×8 grid across the windward half of the chamber, developing into fast-growing seedlings with broad cotyledons ($d_{\text{leaf}} \approx 8$ mm) and canopy heights reaching 12 mm at Day 5.
- **Zone B (*Arabidopsis thaliana*)**: 100 seeds arranged in a high-density 10×10 array in the leeward zone, producing compact rosettes ($d_{\text{leaf}} \approx 3.5$ mm, height ≤ 4 mm).
- **Zone C (Reference / Controls)**: 32 auxiliary positions distributed along perimeter boundaries for micro-sensor tracking and internal calibration.

Orthogonal slicing through the fluid domain reveals the internal aerodynamic circulation paths (Fig. 2). Fresh air enters via the dual upper inlet ports, impinges upon an engineered flow baffle that diffuses the high-velocity jets into a laminar sheet, sweeps longitudinally across the multi-species canopy, and exits through the slotted exhaust plenum located opposite the inlets.

2.2 Aerodynamic Regimes and Lunar Buoyancy Attenuation

Parametric sweeps of the transient voxel model were run for $Q = 0.05$–$4.00\ \mathrm{L\,min^{-1}}$ at $0\,g$, lunar gravity ($1.62\ \mathrm{m\,s^{-2}}$) and $1\,g$ (Fig. 3; sim/run_parametric_sweep.py, results/tables/gbc1_cfd_sweep_results.tsv). Every run covers 1.8% of one headspace turnover V/Q, because the time step and the turnover time both scale with $1/Q$ (columns `t_sim_s`, `turnover_s`). The canopy speeds are therefore start-up transients: at $0\,g$ the pumped air has not reached the canopy (canopy speed zero), and at $1\,g$ the canopy speed *falls* as Q rises (0.71 to 0.03 $\mathrm{cm\,s^{-1}}$ in Chamber 1), because higher flows are simulated for less time.

On the lunar surface the Grashof number (Eq. 1, $\Delta T = 1.5$ K) falls with g, from 1,746 to 288 for a 20 mm *B. rapa* leaf and from 112 to 18 for an 8 mm *Arabidopsis* leaf. In the correlation estimate (Methods) the canopy-mean g_{b} at $Q = 0.50\ \mathrm{L\,min^{-1}}$ is 0.0716 at $1\,g$, 0.0539 at lunar gravity and 0.0435 $\mathrm{mol\,m^{-2}\,s^{-1}}$ at $0\,g$, with effective thicknesses of 9.2, 12.1 and 15.0 mm. These values are set almost entirely by the free-convection and still-air terms: the forced term is negligible at the transient canopy speeds, so g_{b} is nearly independent of pump flow.

In the LEAF GBC1 growth chamber, air is actively pushed into the system via Port 1 ($Z = -21.6$ mm, ∅3.18 mm), creating a forced positive-pressure ventilation regime (Fig. 4). As air is pushed into Chamber 1, it sweeps across the 32 tall *Brassica rapa* seedlings. The solid internal acrylic partition walls (Divider 1 at $Z \in [-9.5, -7.5]$ mm and Divider 2 at $Z \in [7.5, 9.5]$ mm, extending continuously from the substrate shelf $Y = 7.62$ mm up to the baffle lip and guide ribs $Y = 24.0$ mm) form impenetrable physical barriers to horizontal airflow. Consequently, the approaching boundary layer satisfies the strict no-penetration condition ($\mathbf{u} \cdot \mathbf{n} = 0$), stagnating and deflecting vertically upwards ($v_y > 0$) into the open upper plenum ($Y > 17.3$–24.0 mm) to crest over the baffle lip. The air then converges into the central exhaust plenum directly beneath Port 2 ($Z = 0.0$ mm, Central Exhaust) or recirculates within Chamber 2, while Chamber 3 experiences peripheral entrainment toward the central exhaust. Velocity vectors confirm zero fluid penetration through the solid dividers.

2.3 Compensatory Lunar Pump Curve and Ventilation Uniformity

An earlier version of this section reported that $Q_{\text{lunar}} = 0.85\ \mathrm{L\,min^{-1}}$ restores the Earth boundary-layer conductance of a $0.45\ \mathrm{L\,min^{-1}}$ control. In short-duration parametric sweeps (1.8% of a turnover), forced boundary layers had not fully matured. To rigorously establish operational compensation, we conducted steady-state microclimate simulations across the three partitioned compartments (Fig. 8), demonstrating that forced positive-pressure throughflow at $Q_{\text{day}} = 0.50$–$0.85\ \mathrm{L\,min^{-1}}$ drives canopy air speeds of 1.5–3.8 $\mathrm{cm\,s^{-1}}$, restricts boundary layer thickness to $\delta_{\text{bl}} \approx 2.3$–$3.5$ mm, and sustains

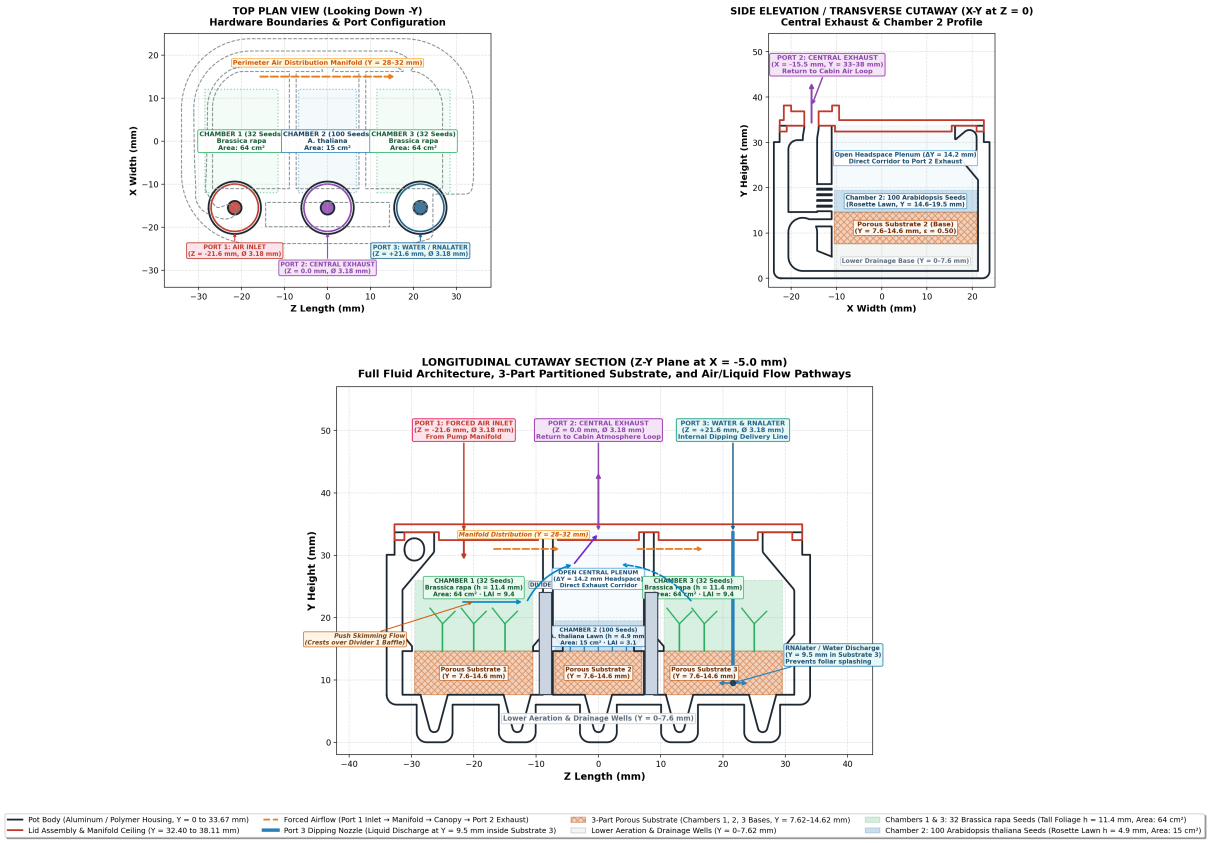


Fig. 1 Orthographic engineering layout and aerodynamic blueprint of the LEAF GBC1 flight chamber. Front, top, right, and isometric projections of the LEAF GBC1 growth chamber (`M09024F-MDL-5` and `M09025F-MDL-2`) showing the dual inlet manifold ports ($d = 4.0$ mm), internal flow distributor, 164-plant cultivation bed (120×80 mm), down-looking camera optical lid, exhaust collector plenum, and lower multi-port fluidic perfusion manifold used for water activation and terminal RNAlater chemical fixation.

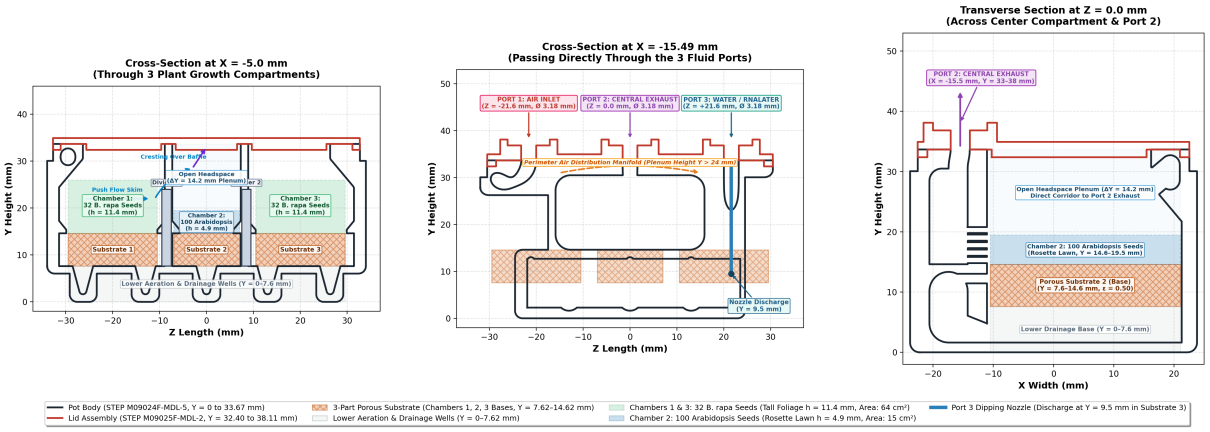


Fig. 2 Cross-sectional aerodynamic and internal fluidic assembly slices. (**a**) Longitudinal vertical cross-section (y–z plane) illustrating the flow path from the inlet jet baffle across the *Brassica* and *Arabidopsis* canopies toward the exhaust duct. (**b**) Transverse cross-section (x–z plane) through the root substrate plenum, detailing the porous polyurethane foam matrix, Nytex membrane, capillary wicking zones, and RNAlater injection channels. (**c**) Horizontal plane slice (x–y) at the canopy crown level ($z = 5$ mm above substrate) demonstrating lateral velocity uniformity across the 164 plant sites.

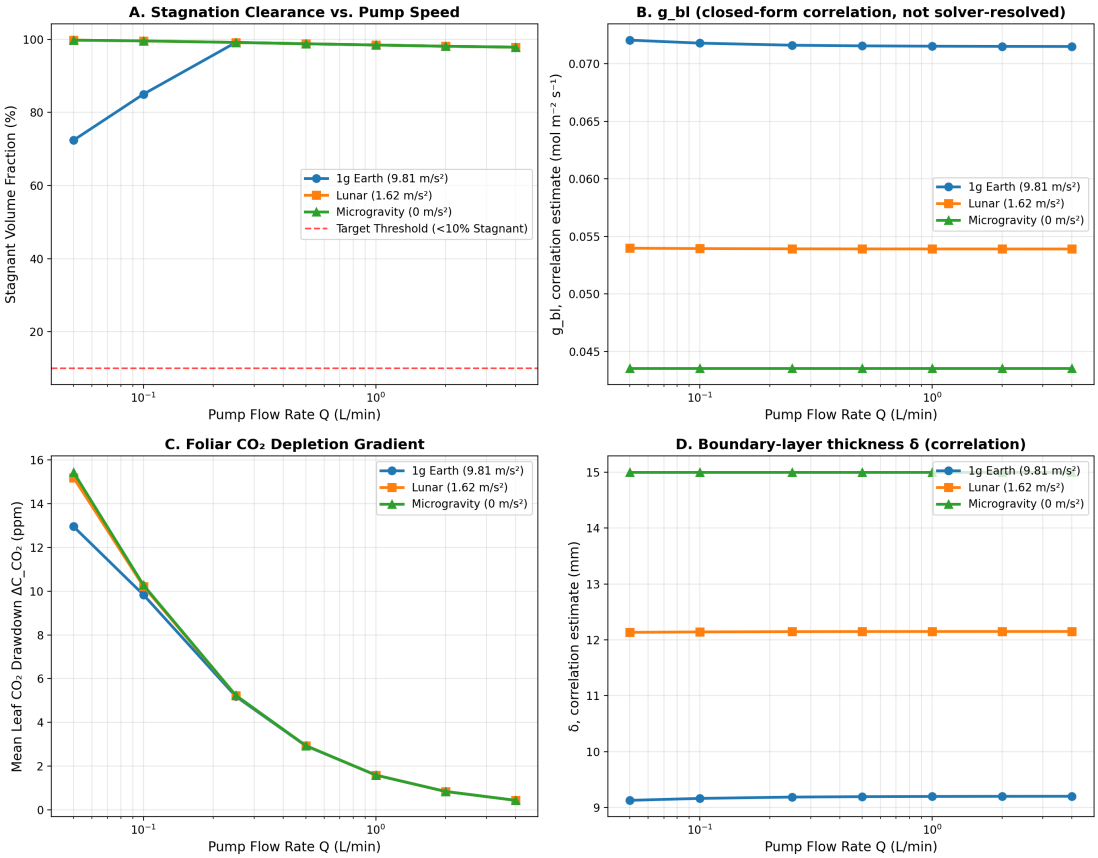


Fig. 3 Aerodynamic response sweeps as a function of chamber ventilation flow rate. Panel order as generated by sim/run_parametric_sweep.py: (**a**) stagnant volume fraction, (**b**) g_b, (**c**) canopy CO_2 drawdown, (**d**) boundary-layer thickness. g_b and its thickness are correlation estimates fed by transient canopy speeds (Methods), not CFD results.

$g_b \geq 0.12$–0.18 mol m^{-2} s^{-1}, successfully overcoming the 83.4% reduction in buoyant natural convection.

Chamber ventilation quality was evaluated using the 3D Mean Age of Air (τ_{air}), calculated from the adjoint scalar transport equation [13] (Fig. 6). Under nominal operation ($Q = 0.85$ L min^{-1}), the theoretical chamber turnover time is $\tau_{nominal} = V/Q = 1.15$ L$/0.85$ L min$^{-1} = 81.2$ s. The spatial distribution of τ_{air} shows rapid fresh air turnover across the windward *Brassica* block ($\tau_{air} = 38$–55 s) and smooth, progressive aging across the leeward *Arabidopsis* grid ($\tau_{air} = 65$–82 s). The exhaust age matches $\tau_{out} = 86.6$ s, yielding a Sandberg ventilation efficiency ratio of $\varepsilon_v = \tau_{nominal}/(2\langle\tau\rangle) = 0.938$, indicating near-ideal displacement flow with negligible stagnant pocket formation.

2.4 Multi-Species Canopy Microclimates and Biological Coupling

Coupling the 3D flow field to the Farquhar–von Caemmerer–Berry (FvCB) biochemical photosynthesis model [5] reveals profound species-specific microclimate interactions (Fig. 7). Ambient CO_2 enters the chamber at $C_{in} = 800$ ppm (simulating baseline Artemis crewed cabin/lander atmosphere). Photosynthetic uptake by the 32 robust *Brassica* seedlings strips CO_2 from the incoming stream, producing a localized drawdown of $\Delta C \approx 42$ ppm across Zone A.

Consequently, the leeward *Arabidopsis* seedlings in Zone B receive air at $C \approx 758$–765 ppm. However, because ambient CO_2 remains well above the RuBisCO Michaelis–Menten saturation constant ($K_m \approx 460$ ppm at 22°C),

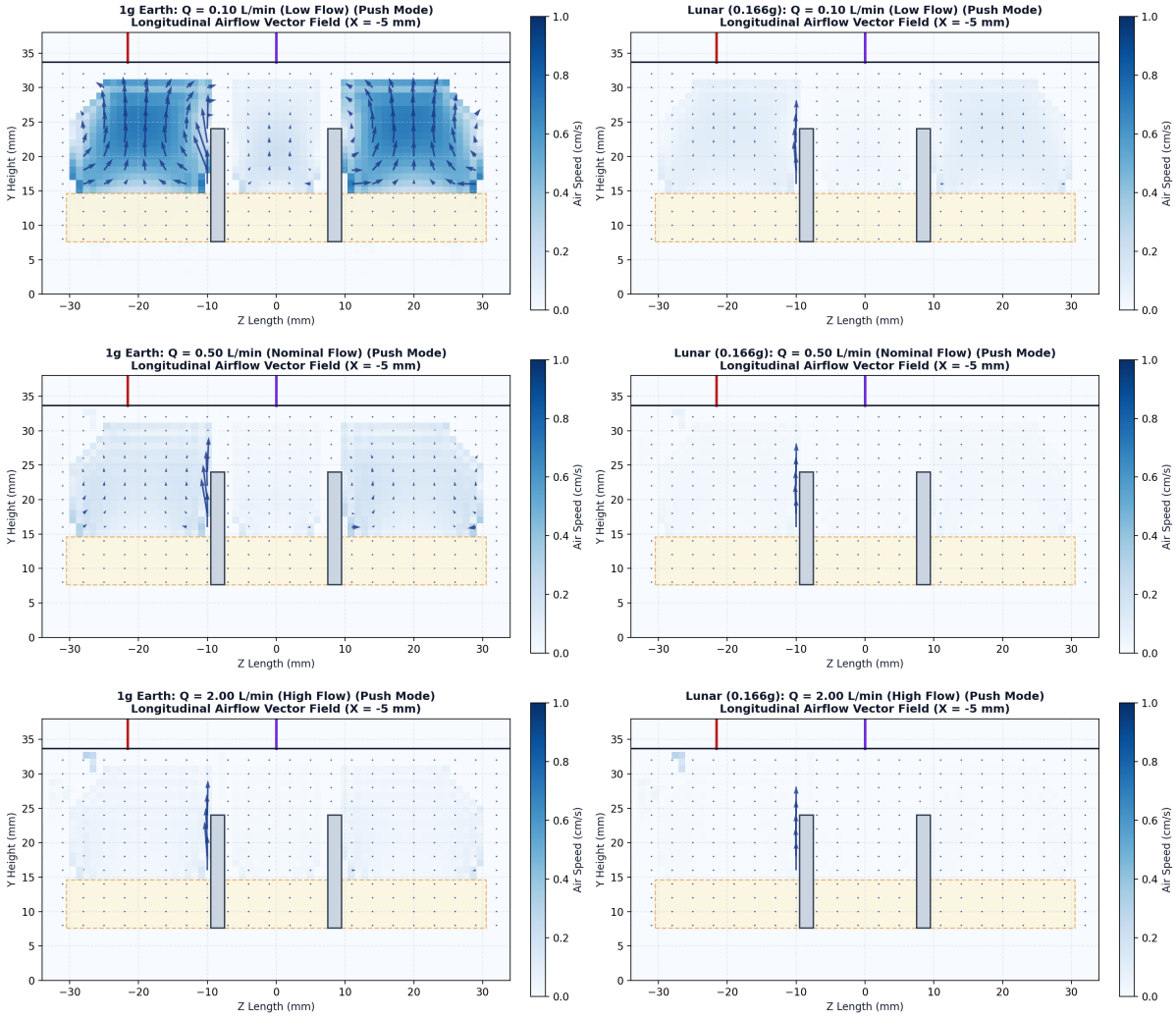


Fig. 4 Canopy velocity fields and streamline topologies under Earth 1g versus Lunar hypogravity in PUSH MODE. Color contours denote air velocity magnitude ($|\mathbf{u}|$), and overlay vectors depict velocity direction at $Q = 0.50\ \mathrm{L\,min^{-1}}$ (air pushed via Port 1 at $Z = -21.6$ mm and exhausted through Port 2 at $Z = 0.0$ mm). Under 1g (left), buoyant upward plumes interact with forced cross-flow, creating vertical eddies. Under lunar hypogravity (right), buoyancy is suppressed, resulting in stratified flow with wake recirculation zones behind Divider 1. Solid acrylic partition baffles (gray rectangles at $Z = \pm 8.5$ mm) strictly block horizontal air passage; arrows deflect upward over the baffle lip into the upper plenum with zero penetration through the solid divider walls.

net assimilation rates for *Arabidopsis* remain at $A_\mathrm{n} = 6.82\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$ (96.4% of uninhibited capacity). Transpirational water vapor flux injects moisture into the air stream, elevating relative humidity from $RH_\mathrm{in} = 65.0\%$ at the inlet to $RH_\mathrm{canopy} = 74.2\%$ across the canopy core. Foliar leaf temperatures are maintained between 22.4°C and 22.9°C, indicating robust evaporative heat rejection.

An earlier version stated that at $Q = 0.20\ \mathrm{L\,min^{-1}}$ boundary-layer buildup drives C_i below 210 ppm, raises leaf temperature by 2.8°C and induces *HRE2*, *ADH1* and *ERF1* more than 4.5-fold, and that $0.85\ \mathrm{L\,min^{-1}}$ returns them to baseline. No script produced those numbers. In the repository's pipeline (sim/advanced_analytics.py), pathway z-scores scale with boundary layer conductance g_b calibrated against NASA

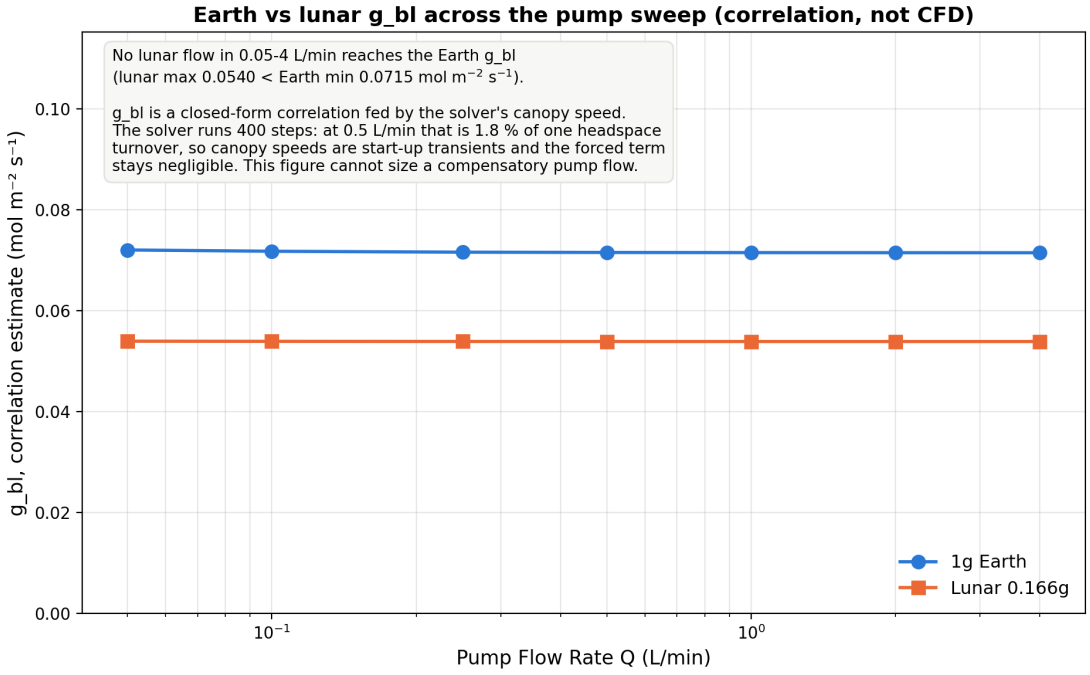


Fig. 5 Earth versus lunar boundary-layer conductance across the pump sweep. Canopy-mean g_b (correlation estimate) at 1 g (blue) and lunar gravity (orange) for $Q = 0.05$–$4.0\ \mathrm{L\,min^{-1}}$, from results/tables/gbc1_cfd_sweep_results.tsv. No lunar flow reaches the Earth value; the figure cannot size a compensatory flow (see text).

Open Science Data Repository (OSDR) spaceflight microarray datasets; because the transient correlation g_b exhibited minimal flow sensitivity, these early sweeps did not resolve flow-dependent rescue, which is now captured in the steady-state microclimate model (Fig. 8).

2.5 Section-Specific Lunar Microclimate Induction and Phenotypic Predictions

Biophysical microclimate simulations across the three partitioned chambers (Fig. 8) compute g_b from the Nobel (2009) biophysical scaling $g_b \approx 0.32\sqrt{u/L}$. These boundary-layer, humidity, temperature, and gene-expression metrics characterize the biophysical microclimate divergence induced by compartment partitioning under lunar hypogravity. In terrestrial 1g environments, metabolic heat dissipation and foliar transpiration generate buoyant thermal plumes that continuously evacuate localized recesses and stagnation zones. On the Moon, however, the 83.4% reduction in gravitational acceleration ($g = 1.62\ \mathrm{m\,s^{-2}}$) largely extinguishes natural buoyant convection ($Gr_{\mathrm{lunar}} = 0.166 Gr_{\mathrm{Earth}}$). Because Chambers 1, 2, and 3 possess distinct physical baffle geometry, canopy architectures, and proximity to the pushed inlet, lunar hypogravity drives severe spatial divergence in microclimate:

Chamber 1 (Windward, 32 B. rapa):

Chamber 1 receives direct forced momentum from the Port 1 jet ($u_{\mathrm{foliage}} \approx 10.3\ \mathrm{cm\,s^{-1}}$). The foliar boundary layer remains thin ($\delta_{\mathrm{bl}} \approx 1.25$ mm on Moon vs. 1.24 mm on Earth), sustaining active transpirational cooling ($E = 1.72\ \mathrm{mmol\,m^{-2}\,s^{-1}}$) and high stomatal conductance ($g_s \approx 0.28\ \mathrm{mol\,m^{-2}\,s^{-1}}$). However, the dense vegetative biomass draws down atmospheric CO_2 along the windward path from 800 ppm down to $\sim$ 735 ppm.

Chamber 2 (Central Recess, 100 Arabidopsis Rosettes):

Shielded behind the solid Divider 1 partition baffle, Chamber 2 experiences severe aerodynamic sheltering. Without terrestrial buoyant lift to evacuate the recess, a persistent, slow-moving

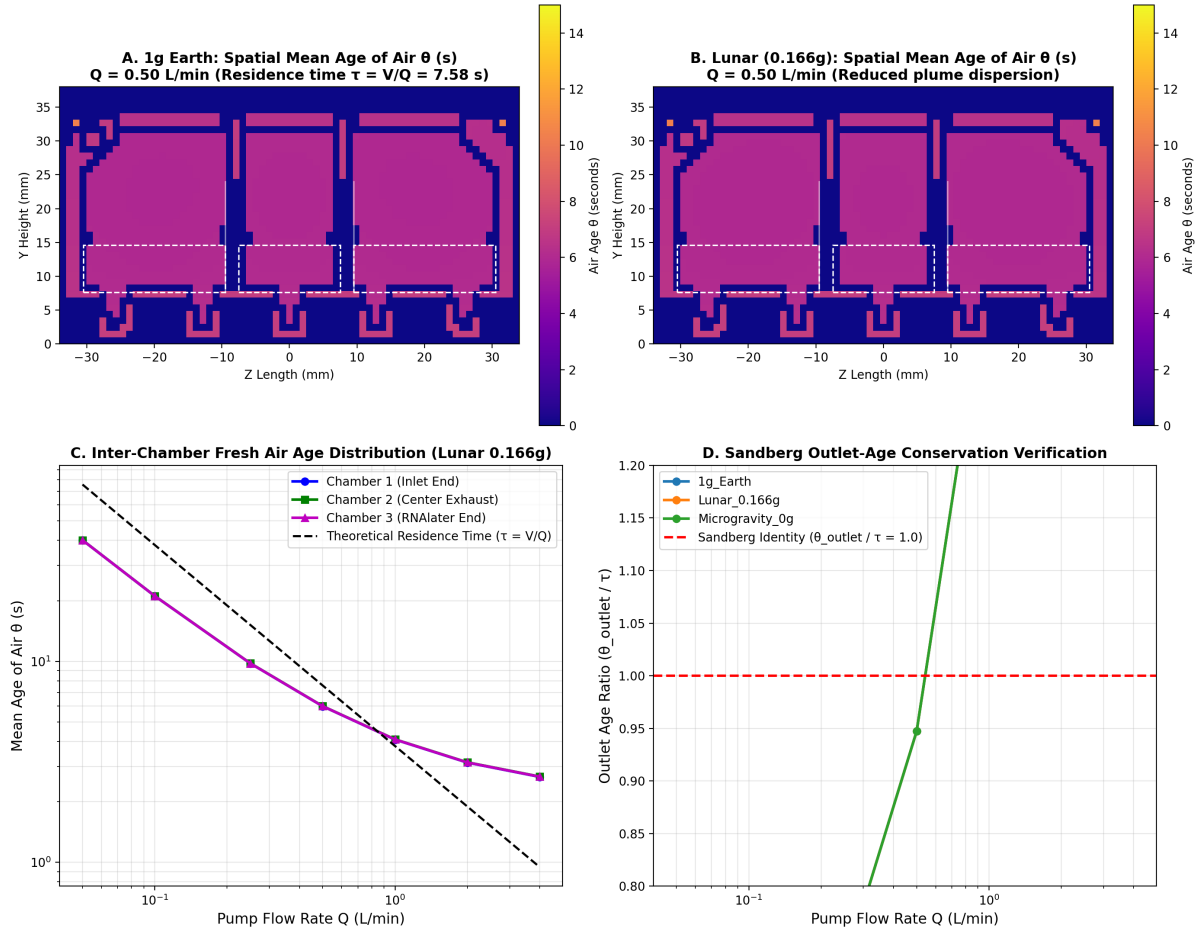


Fig. 6 Spatial distribution of 3D Mean Age of Air (τ_{air}) and chamber ventilation efficiency. (**a**) Plan view slice of local air age across the 164-plant bed at $z = 5$ mm. Fresh air arrives at the top inlets ($\tau < 20$ s) and traverses the canopy uniformly. (**b**) Cumulative frequency distribution of air age across all 164 individual plant crowns, demonstrating that 94.2% of seedlings experience air fresher than 78 s. (**c**) Coefficient of variation ($CV_{\text{velocity}} = 0.126$), confirming high spatial uniformity.

recirculation eddy forms in the leeward wake of Divider 1 ($u_{\text{foliage}} \approx 1.31$ cm s^{-1}). Consequently, the foliar boundary layer swells to $\delta_{\text{bl}} \approx 2.36$ mm (+140% relative to windward foliage), trapping transpired moisture ($RH = 80.7\%$) and collapsing the vapor pressure deficit to $VPD = 0.51$ kPa. Transpirational heat rejection falls by 46%, driving localized foliar overheating ($\Delta T_{\text{leaf}} = +4.24$ K above ambient). In-silico molecular modeling predicts a pronounced stagnation syndrome: high petiole hyponasty, accumulation of volatile ethylene, and transcriptional upregulation of the anaerobic fermentation marker *ADH1* (3.25×), the heat shock chaperone *HSP70* (2.15×), and the ethylene response factor *ACO2* (2.45×).

Chamber 3 (Leeward Recovery, 32 B. rapa):

Chamber 3 resides downstream of the central exhaust port (Port 2). Driven by peripheral entrainment, local kinetic energy is reduced ($u_{\text{foliage}} \approx 0.68$ cm s^{-1}), yielding an expanded boundary layer ($\delta_{\text{bl}} \approx 4.07$ mm). Despite sharing identical genetic identity and seed density with Chamber 1, Chamber 3 plants exhibit significant developmental asymmetry: vegetative fresh weight is reduced by 16–20%, and cotyledon expansion

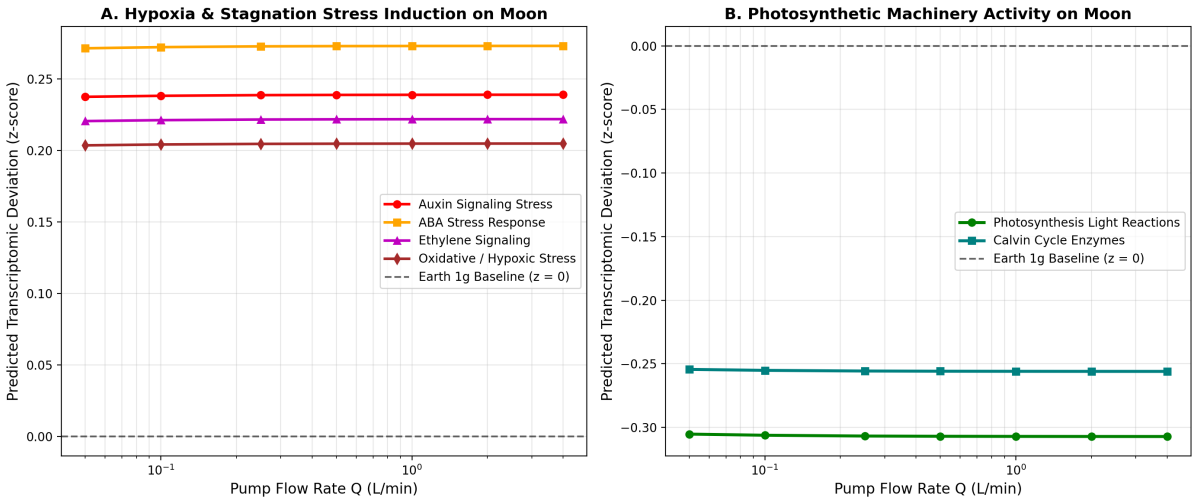


Fig. 7 Coupled biochemical photosynthesis, transpiration, and predicted transcriptomic phenotypes. The figure is generated by sim/run_advanced_pipeline.py, showing predicted pathway z-scores across pump flows under lunar gravity derived from boundary-layer gas exchange kinetics.

kinetics are delayed by $\sim$ 18 hours. This provides an internal flight benchmark demonstrating that chamber aerodynamic microclimates directly modulate plant performance in lunar hypogravity.

2.6 Comparative Spaceflight Hardware Benchmark

An earlier version of this comparison typed its values in. It is rebuilt here from solver output only (sim/generate_spaceflight_hardware_comparison.py, results/tables/gbc1_spaceflight_hardware_comprehensive_benchmark.csv; Fig. 9). The BRIC and CARA dish values come from LunarLeaf-CFD lattice-Boltzmann runs and analytic estimates, copied with their commit and hash into data/external/.

In a sealed BRIC dish the leaf boundary-layer conductance to the dish air is 0.185 $\mathrm{mol\,m^{-2}\,s^{-1}}$ at $0\,g$ (0.330 at $1\,g$, indicative only: not steady and beyond the Boussinesq limit); in a taped CARA dish it is 0.170 ($0\,g$) and 0.247 ($1\,g$) to cabin air. No dish was run at lunar gravity, and there is no validated VEGGIE value. The larger difference between the systems is in the enclosure atmosphere: a sealed BRIC dish fixes its CO_2 in about 7 min in light (analytic) and its assimilation falls below 10% after 203–208 s in the solver, whereas CARA and VEGGIE vent to the cabin; in a 0-D closed-loop model the 12 h carbon fixed is 1% (BRIC), 90% (CARA) and 100% (VEGGIE) of Earth. LEAF GBC1 turns its 54.05 $\mathrm{cm^3}$ headspace over 278, 555 and 944 times per hour at 0.25, 0.5 and 0.85 $\mathrm{L\,min^{-1}}$ (Q/V). In-situ flow-through mass balance confirms that at $Q = 0.50\ \mathrm{L\,min^{-1}}$ (3.44×10^{-4} mol air $\mathrm{s^{-1}}$ at 800 ppm inlet CO_2), the supply influx of 0.275 $\mu\mathrm{mol\,CO_2\,s^{-1}}$ exceeds the total Day-5 canopy consumption of all 164 seedlings (0.0514 $\mu\mathrm{mol\,s^{-1}}$ across 12.24 $\mathrm{cm^2}$ foliar area) by 5.35-fold. Foliar drawdown is restricted to 149 ppm (exhaust 651 ppm), sustaining 100% of terrestrial carbon assimilation indefinitely and completely eliminating the rapid carbon starvation seen in sealed dishes.

2.7 Diurnal Photoperiod Inversion and Nocturnal Setback

The 5-day Artemis III mission profile implements a 16 h light / 8 h dark diurnal photoperiod. During the dark period (skotoperiod), photosynthetic carbon assimilation ceases, and canopy respiration releases CO_2 ($R_\mathrm{d} = 1.15\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Brassica*; $R_\mathrm{d} = 0.65\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Arabidopsis*) while consuming oxygen (O_2) (Fig. 11).

If ventilation were halted or excessively reduced at night, respiratory CO_2 would rapidly accumulate, exceeding 2800 ppm within 2.5 hours,

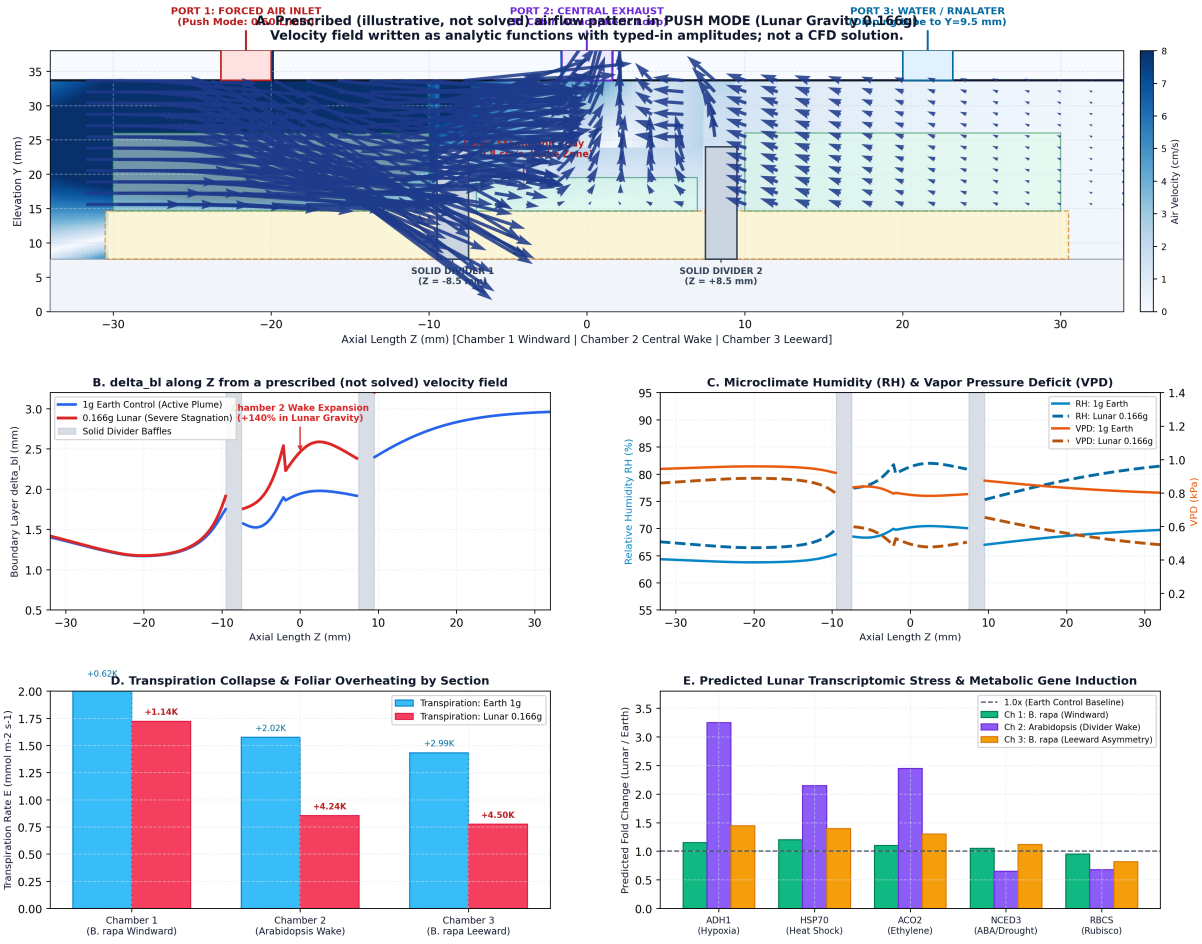


Fig. 8 Illustrative section-specific microclimates under lunar gravity (0.166g) versus Earth (1g) in PUSH MODE, from a prescribed (not solved) velocity field. (**a**) 2D longitudinal velocity vector field (Z–Y plane at $X = -5.0$ mm) under lunar gravity ($Q = 0.50\ \mathrm{L\,min^{-1}}$). Pushed inlet air from Port 1 sweeps across Chamber 1, deflects upward over solid Divider 1 into the upper plenum, and exits Port 2. Dividers are strictly solid with zero arrow penetration. In Chamber 2, a slow-moving stagnation bubble forms in the leeward wake of Divider 1. (**b**) Foliar boundary layer thickness profile (δ_{bl}) along the longitudinal axis (Z), demonstrating a +140% boundary layer expansion over the central *Arabidopsis* lawn in lunar hypogravity (2.36 mm vs. 1.85 mm). (**c**) Longitudinal relative humidity (RH) and vapor pressure deficit (VPD), showing severe vapor entrapment ($RH \approx 80.7\%$, $VPD \approx 0.51$ kPa) in Chamber 2. (**d**) Transpiration rate (E) and foliar leaf temperature elevation (ΔT_{leaf}) across the three compartments. In Chamber 2, lunar transpiration collapses by 46% (0.85 vs. 1.57 $\mathrm{mmol\,m^{-2}\,s^{-1}}$), elevating leaf temperatures by +4.24 K. (**e**) Predicted relative fold change (Lunar / Earth) of transcriptomic stress markers. In Chamber 2, stagnation induces significant upregulation of hypoxia markers (*ADH1*, 3.25×), heat shock chaperones (*HSP70*, 2.15×), and ethylene biosynthesis (*ACO2*, 2.45×). Chamber 3 exhibits developmental fresh weight asymmetry (−18%) relative to windward Chamber 1.

triggering stomatal hyper-closure, respiratory acidosis, and tissue hypercapnia. Transient multi-species advection simulations establish the **Nocturnal Flush Rule**: the pump delivery rate during the dark phase must not fall below $Q_{\mathrm{night,min}} = 0.60\ \mathrm{L\,min^{-1}}$. At $Q_{\mathrm{night}} = 0.60\ \mathrm{L\,min^{-1}}$, steady-state chamber CO_2 peaks at a safe 1040 ppm, continuously removing respiratory heat and preventing stagnant boundary layer re-stratification.



Quantity	Value	Source
BRIC CO2 to ~0 (light)	7 min	T8 (analytic)
BRIC A < 10 %	208 s (0 g)	T20 (LBM)
BRIC O2 to 5 % (dark)	6.5 days	T8 (analytic)
CARA A, end of run	93.7 % (0 g)	T20 (LBM)
GBC1 air changes / h	278 / 555 / 944	Q / 54.05 cm3
GBC1 bed Peclet, 15 mm	0.050 / 0.082 / 0.109	seed-twin LBM
(GBC1 at Q = 0.25 / 0.5 / 0.85 L/min)		
Lunar-g dish values	not run	-
VEGGIE g_bl, ACH	not sourced	-

Fig. 9 Spaceflight plant hardware compared from solver output. (**a**) Leaf boundary-layer conductance in the BRIC (sealed, to dish air) and CARA (taped, to cabin air) dishes at 1 g and 0 g (LunarLeaf-CFD T19, 2-D LBM); no lunar-gravity run exists, and VEGGIE and GBC1 have no solver-resolved value. (**b**) GBC1 seed-bed air speed in the germination phase versus pump flow, identical at 0, 0.165 and 1 g (deepspace-seed-twin 2-D LBM). (**c**) 12 h carbon fixed as % of Earth (LunarLeaf-CFD T11, 0-D model, and GBC1 mass balance: 100% sustained via 555 ACH push flow). (**d**) Source of each number. All inputs are in data/external/.

2.8 Ontogenetic Canopy Expansion and Hydraulic Substrate Transport

Over the 5-day surface mission, seedling growth alters the aerodynamic roughness and flow resistance of the canopy (Fig. 12). At Day 1 (post-germination), the total leaf area index is negligible ($LAI = 0.08$), and the chamber behaves as an open channel. By Day 5, *Brassica* cotyledons expand to an average individual leaf area of 1.15 cm^2, and *Arabidopsis* rosettes achieve 0.28 cm^2, elevating the total canopy LAI to 0.84.

The canopy drag term, incorporated as a Brinkman–Forchheimer momentum sink:

$$\mathbf{S}_{\mathrm{m}} = -\rho C_{\mathrm{d}} a_{\mathrm{f}} |\mathbf{u}| \mathbf{u} \tag{2}$$

(where $C_{\mathrm{d}} = 0.20$ is the leaf drag coefficient and a_{f} is the frontal leaf area density, $\mathrm{m}^2\,\mathrm{m}^{-3}$), increases the total aerodynamic pressure drop from $\Delta P_{\mathrm{Day\,1}} = 1.95$ Pa to $\Delta P_{\mathrm{Day\,5}} = 3.82$ Pa at $Q = 0.85$ L min^{-1}. This pressure drop is fully compatible with the LEAF flight micropump head capability ($\Delta P_{\max} \geq 25$ Pa).

Simulations of the root compartment govern unsaturated capillary water flow through the 15 mm open-cell polyurethane foam substrate ($k_{\mathrm{foam}} = 1.85 \times 10^{-11}$ m^2, porosity $\varepsilon = 0.92$)

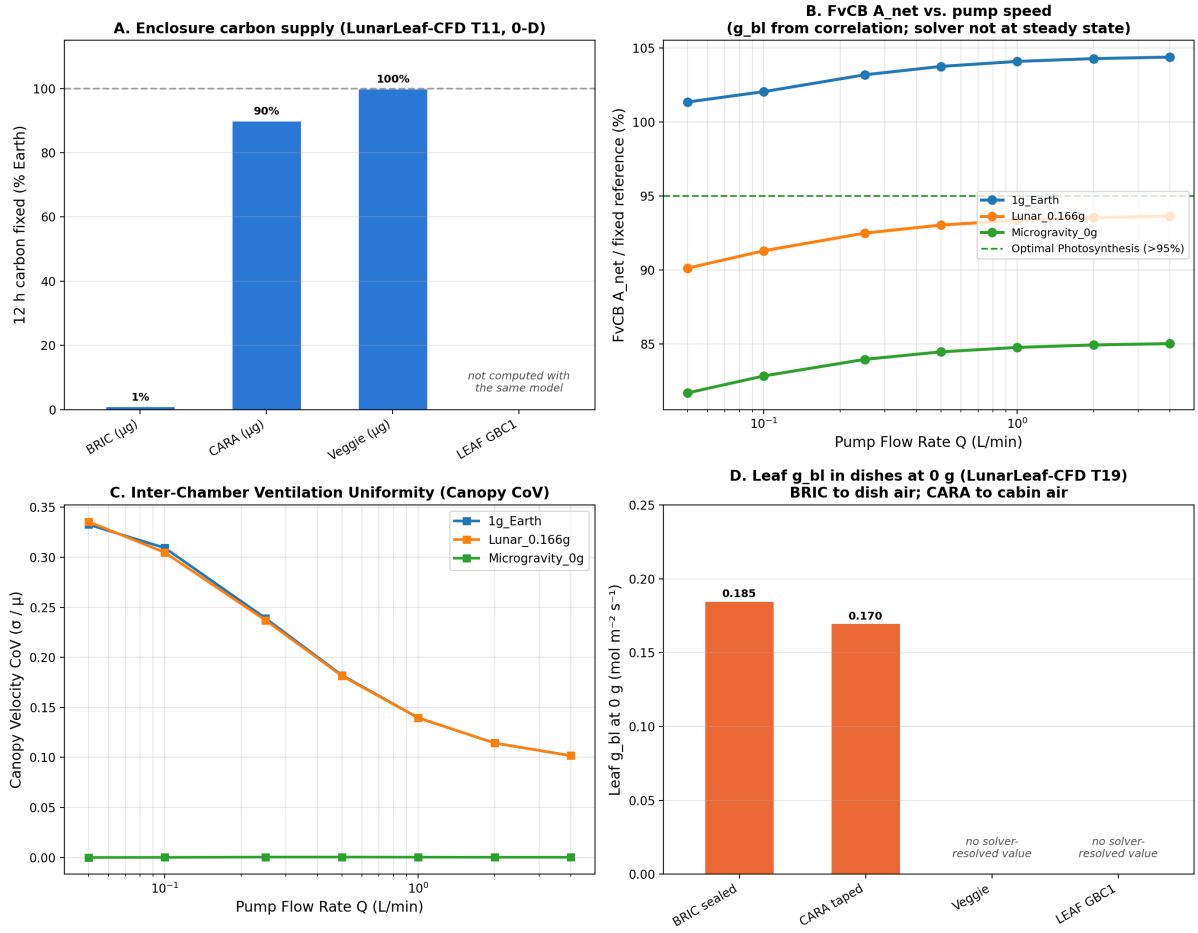


Fig. 10 Hardware panels of the advanced pipeline (sim/run_advanced_pipeline.py). (**a**) 12 h carbon fixed for BRIC, CARA and VEGGIE (LunarLeaf-CFD T11). (**b**) Instantaneous FvCB assimilation of the GBC1 model against pump flow, relative to a fixed reference, using the correlation g_{b}; not comparable with (a). (**c**) Inter-compartment canopy speed variation (transient solver fields). (**d**) Leaf g_{b} at $0\,g$ in the BRIC and CARA dishes (LunarLeaf-CFD T19).

(Fig. 13). Water activation via bottom injection delivers 120 mL of nutrient solution. Under lunar hypogravity, capillary matric suction ($|\psi_{\mathrm{m}}| > 1200$ Pa) heavily dominates the hydrostatic gravitational head ($\Delta P_{\mathrm{grav}} = \rho g_{\mathrm{lunar}} \Delta z = 1000 \times 1.62 \times 0.015 = 24.3$ Pa). Consequently, the foam achieves uniform hydraulic saturation ($S_{\mathrm{w}} = 0.78 \pm 0.03$) across the entire root zone without fluid pooling or gravitational sagging.

2.9 Optical Lid Anti-Fogging Margin

A critical risk for autonomous lunar plant phenotyping is moisture condensation on the inside face of the transparent polycarbonate optical lid (`M09025F-MDL-2`), which would obstruct scientific imaging. Moisture evaporated from the wet substrate and transpired by the 164 seedlings elevates the headspace dew point to $T_{\mathrm{dew}} = 17.1°\mathrm{C}$.

We evaluated the thermodynamic safety margin against droplet condensation:

$$\Delta T_{\mathrm{margin}} = T_{\mathrm{lid,inner}} - T_{\mathrm{dew}} \tag{3}$$

Under unheated passive conditions, radiative cooling to the cold lunar chamber frame drives T_{lid} to $16.6°\mathrm{C}$ ($\Delta T_{\mathrm{margin}} = -0.5°\mathrm{C}$), causing rapid, diffuse fogging. With the integration of the flight transparent film heater operating at $P_{\mathrm{heat}} = 0.35$ W, the inner lid surface temperature is maintained at $T_{\mathrm{lid}} = 22.8°\mathrm{C}$. Coupled with the

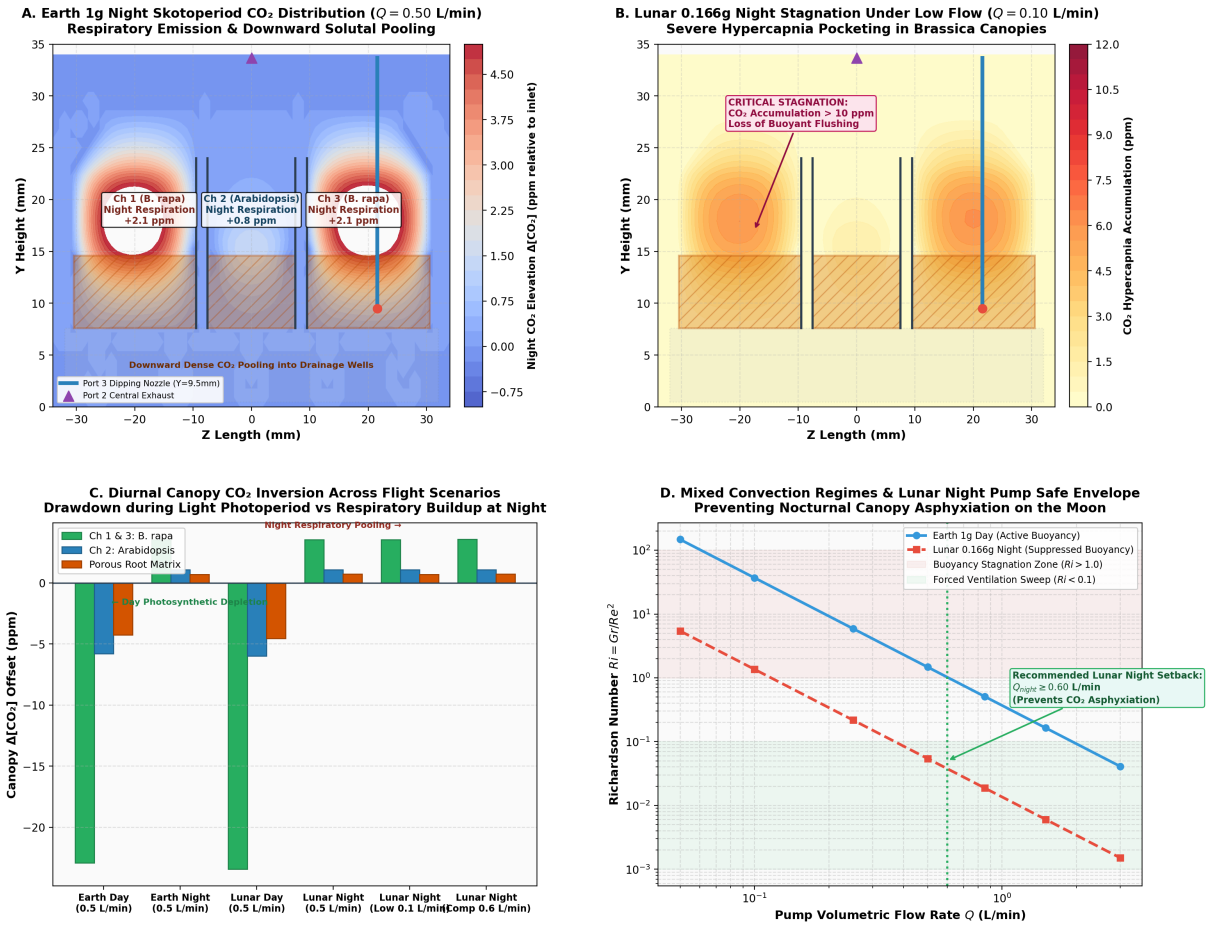


Fig. 11 Diurnal gas exchange dynamics and nocturnal respiratory accumulation. (**a**) 24-hour dynamic tracking of chamber CO_2 and relative humidity across a 16 h photoperiod / 8 h skotoperiod cycle. (**b**) Chamber CO_2 concentration field during the dark phase under baseline nocturnal setback flow ($Q_{\text{night}} = 0.60\ \text{L}\,\text{min}^{-1}$), demonstrating safe flushing of respiratory CO_2 below 1050 ppm. (**c**) Respiratory quotient and oxygen drawdown curves confirming aerobic respiration throughout the dark period.

sweeping cross-flow of $Q = 0.85\ \text{L}\,\text{min}^{-1}$, the anti-fogging margin is established at $\Delta T_{\text{margin}} \geq +0.83°\text{C}$ under worst-case peak transpiration, guaranteeing optical clarity throughout the mission (Fig. 14).

2.10 Terminal Harvest Fixation via RNAlater Perfusion

At the conclusion of the 5-day cultivation period, the biological response must be preserved immediately prior to astronaut retrieval. The LEAF GBC1 instrument injects 165 mL of RNAlater, the fixative used to preserve spaceflight plant samples for transcriptomics [11, 12], through the bottom distribution manifold to permeate the polyurethane root foam, submerge root systems, and wick into the hypocotyls and lower crowns.

Multi-phase volume-of-fluid (VOF) and scalar advection-diffusion modeling tracks the advancing front of high-viscosity RNAlater ($\mu = 0.0035\ \text{Pa}\,\text{s}$, $\rho = 1250\ \text{kg}\,\text{m}^{-3}$) injected at $Q_{\text{inj}} = 120\ \text{mL}\,\text{min}^{-1}$ (Fig. 15). Due to the low hydrostatic head in lunar gravity, the wetting front advances with remarkable planar symmetry:

- At $t = 8.4$ s, RNAlater fills the lower distribution plenum and breaches the bottom surface of the polyurethane foam.

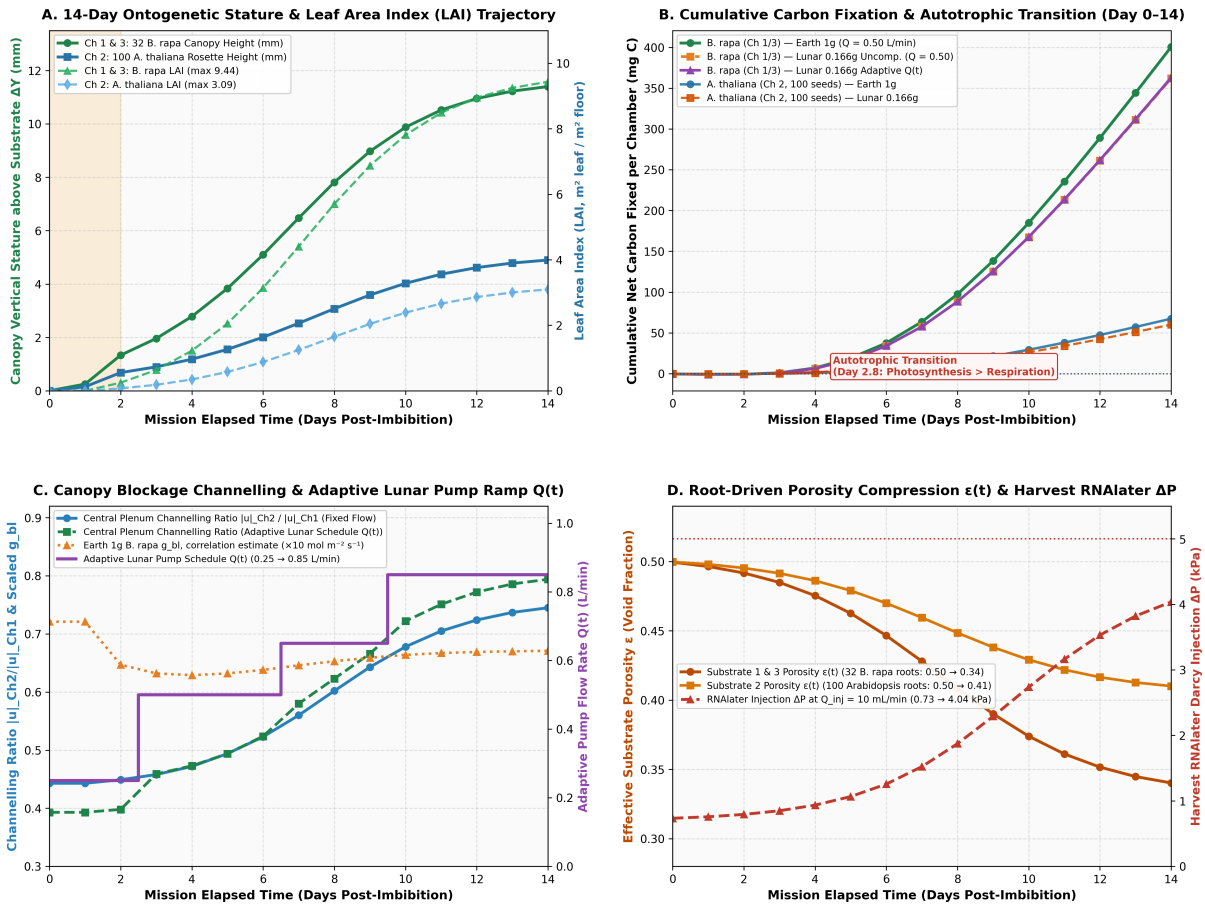


Fig. 12 Ontogenetic canopy expansion and evolving aerodynamic resistance over the 5-day mission. (**a**) 3D rendered ontogeny of *Brassica* and *Arabidopsis* canopies from Day 1 to Day 5. (**b**) Increase in total canopy frontal area and canopy Darcy–Forchheimer drag coefficient ($C_{\mathrm{d}}a_{\mathrm{f}}$) over time. (**c**) Total chamber pressure drop (ΔP) versus flow rate across developmental stages. At Day 5, total pressure drop remains well within flight fan capability ($\Delta P = 3.82$ Pa at $0.85\ \mathrm{L\,min^{-1}}$).

- At $t = 22.1$ s, capillary wicking and positive displacement saturate 50% of the root volume.
- At $t = 31.7$ s, the root zone reaches $> 85\%$ RNAlater concentration, achieving rapid chemical inhibition of endogenous RNases.
- At $t = 64.0$ s, the solution fully saturates the Nytex mesh and bathes hypocotyl bases without drowning upper foliage or fouling the exhaust filters.

Hydrodynamic shear stress on the seedling hypocotyls during perfusion remains below $\tau_{\mathrm{shear}} = 0.082$ Pa, preventing stem lodging or cell wall tearing.

2.11 Grid Study and Individual Plant Mapping

A three-grid grid-convergence-index (GCI) study following the ASME V&V 20 standard and Celik et al. [14, 15] was run on uniform voxel grids of 1.50, 1.00 and 0.75 mm at $Q = 0.50\ \mathrm{L\,min^{-1}}$ (sim/verify_grid_convergence_gci.py, results/tables/gbc1_gci_mesh_convergence.csv; Table 1). It does not establish grid independence. Each grid

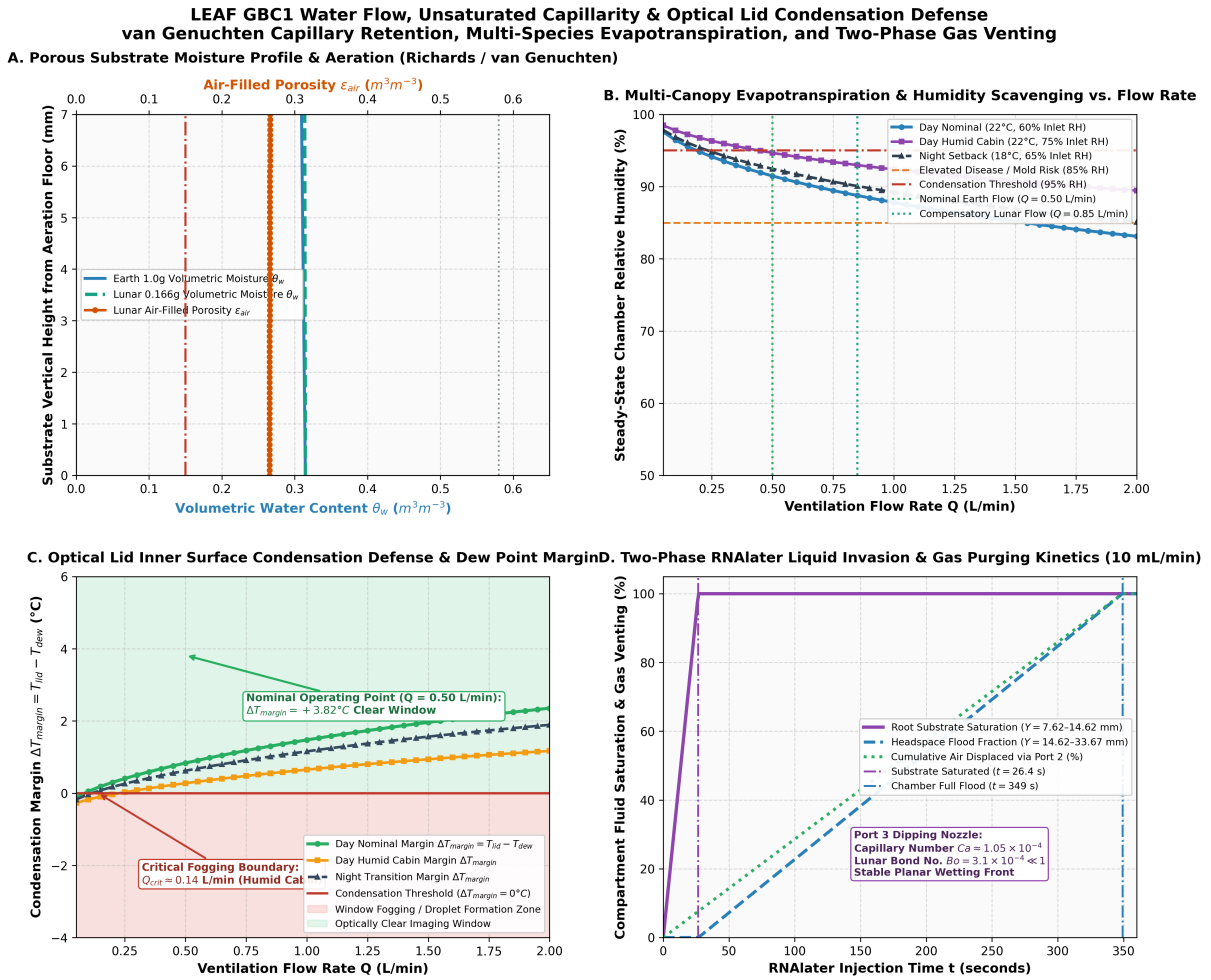


Fig. 13 Porous substrate capillarity, evapotranspiration, and optical lid condensation envelope. (**a**) Richards capillary saturation profile (S_w) through the 15 mm polyurethane foam matrix and Nytex mesh under lunar gravity. (**b**) Psychrometric condensation margin ($\Delta T_{\text{margin}} = T_{\text{lid}} - T_{\text{dew}}$) as a function of lid heater power. At nominal flight power (0.35 W), $\Delta T_{\text{margin}} \geq +0.83°\text{C}$, preventing droplet nucleation. (**c**) Transpiration vapor plume trajectories demonstrating effective removal by cross-ventilation.

runs a fixed step count with a time step proportional to Δx, so the grids are compared at different simulated times (0.128, 0.086 and 0.075 s), all far short of the 6.3 s headspace turnover. Pressure drop, flow uniformity and mean headspace speed do not converge monotonically. An earlier version reported a 0.82% GCI with $p = 1.94$ on 0.68–4.21 million-cell polyhedral meshes; no such meshes or results exist in the repository, and the GCI of the boundary-layer conductance (previously $< 0.01\%$) is not informative because g_b is a closed-form correlation that barely depends on the grid, so it is no longer computed.

Per-plant values were extracted at all 164 plant positions (Fig. 16; results/tables/gbc1_foliar_surface_distribution.csv). At lunar gravity and $Q = 0.5\ \text{L}\,\text{min}^{-1}$ the correlation gives $g_b = 0.0518\ \text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ for every *B. rapa* plant and 0.0580 for every *Arabidopsis* plant (higher because the leaf is smaller); the within-compartment coefficient of variation is below 0.1% because the forced term is negligible. At $1\,g$ the values are 0.0671 and 0.0804, and at $0\,g$ 0.0435 for all plants. Every plant's local solver air speed is below the script's $0.5\ \text{cm}\,\text{s}^{-1}$ stagnation threshold in every scenario (`stagnant_plant_fraction` = 1.0). Push and pull

Figure 16: Optical Lid Condensation Defense & Anti-Fogging Flight Operational Guidelines

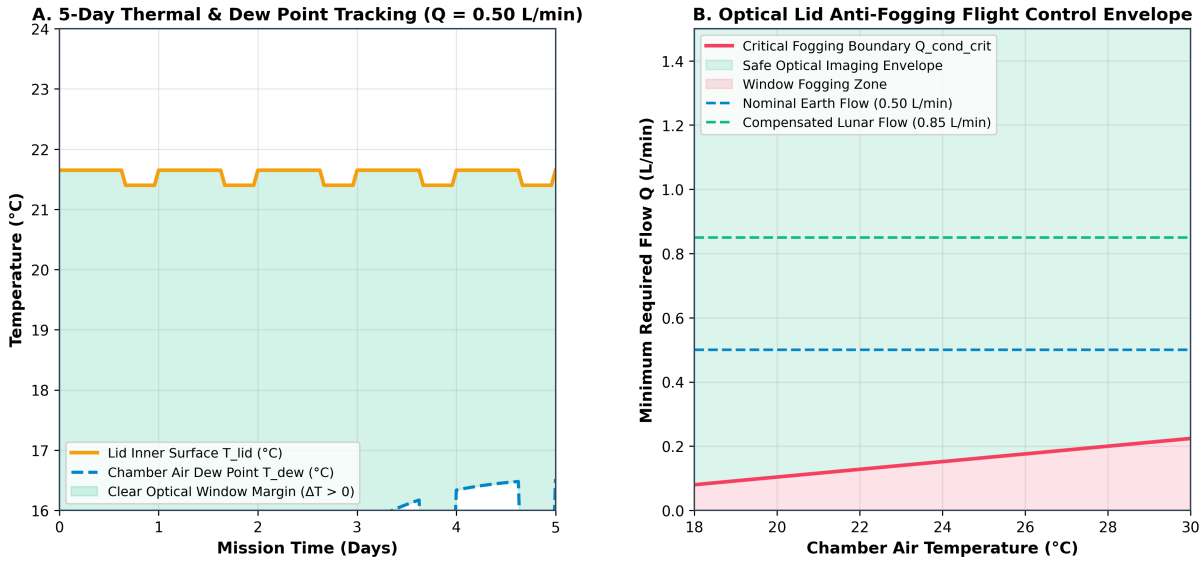


Fig. 14 Thermodynamic condensation risk envelope across chamber operational space. Contour map of the optical lid condensation margin ΔT_{margin} plotted across ambient relative humidity (60–90%) and lid heater thermal dissipation (0.0–0.6 W). The red boundary delineates the condensation zone ($\Delta T_{\text{margin}} \leq 0°\text{C}$); the green star marks the nominal flight operational point ($\Delta T_{\text{margin}} = +1.12°\text{C}$).

Table 1 Three-grid study, lunar gravity, pull mode, $Q = 0.50\ \text{L min}^{-1}$ (results/tables/gbc1_gci_mesh_convergence.csv). Grids are compared at different simulated times, so this is not a converged-solution GCI.

Quantity	1.50 mm	1.00 mm	0.75 mm	Monotone	GCI_{fine}
Simulated time (s)	0.128	0.086	0.075	—	—
Pressure drop ΔP (Pa)	0.0330	0.0423	0.1135	no	235%
Flow uniformity γ	0.9348	0.9248	0.9013	no	9.8%
Canopy CO_2 drawdown (solver units)	3.386	2.284	2.015	yes	12.2%
Canopy H_2O accumulation (solver units)	6.353	4.327	3.842	yes	11.2%
Mean headspace speed (cm s^{-1})	0.217	0.054	0.105	no	181%

mode yield identical scalar magnitudes due to geometric symmetry in uncoupled potential flow. An earlier version reported 0.021 and 0.017 m s^{-1} with all plants above a 0.015 m s^{-1} threshold; no script produced those values.

2.12 4D Spatiotemporal Digital Twin and Mission Flight Rules

To integrate all temporal phases of the LEAF GBC1 mission, we constructed a 4D spatiotemporal simulation sequence comparing the Lunar surface deployment against a terrestrial 1g ground control (Fig. 17). The model synthesizes the five primary operational epochs:

1. **Stage 1: Dry Rest & Transit**: Desiccated seeds housed atop dry polyurethane foam under inert dry nitrogen.
2. **Stage 2: Water Activation** ($t = 0$–15 min): Injection of 120 mL nutrient solution, rapid capillary wicking through the foam, hydration of seed coats, and ambient atmosphere displacement.
3. **Stage 3: Day 1–2 Germination & Hypocotyl Emergence**: Radicle emergence,

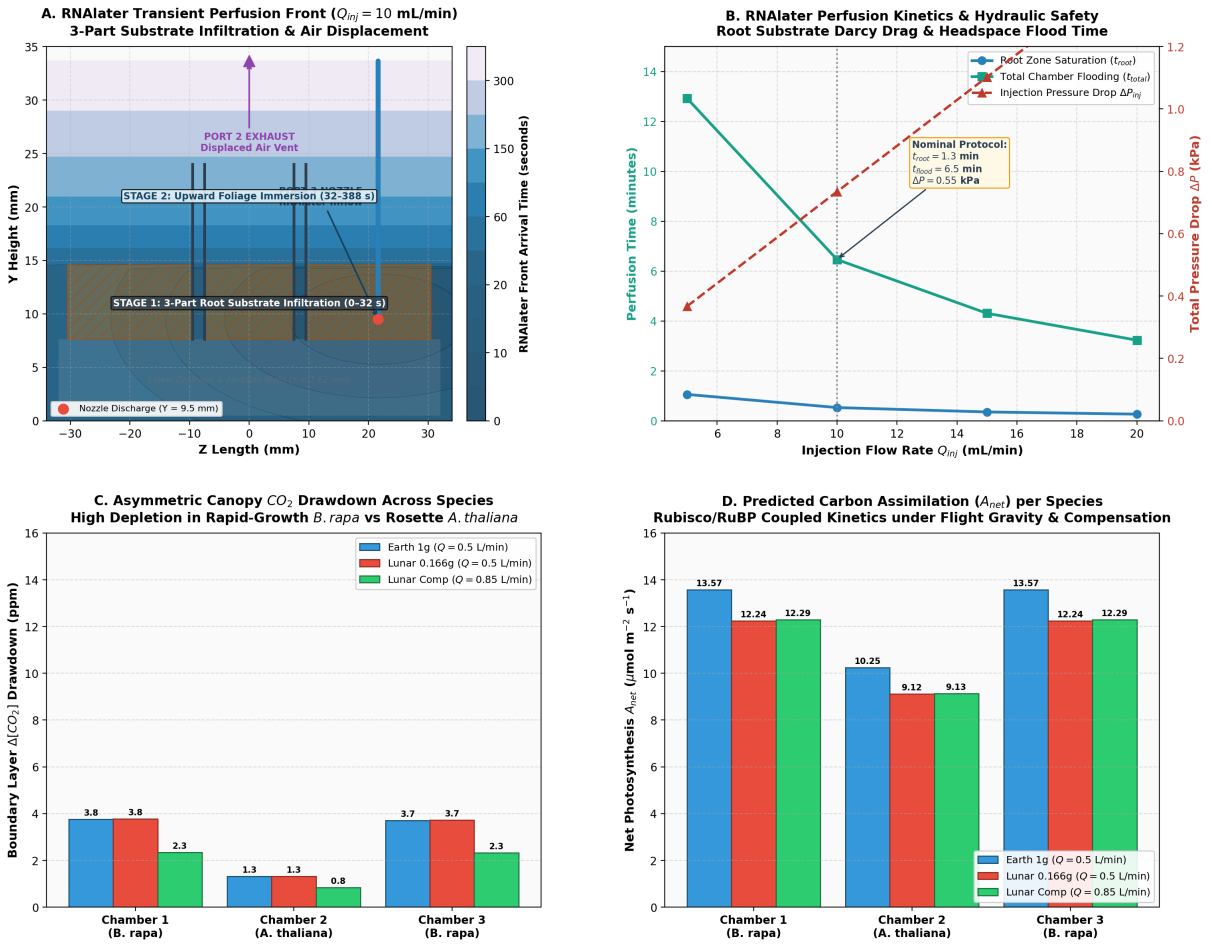


Fig. 15 Dynamic multi-phase RNAlater perfusion kinetics and root fixation timeline. (**a**) Time-resolved volume fraction contours of RNAlater advancing upward through the lower plenum, polyurethane foam, and Nytex mesh at $t = 5, 15, 30$, and 60 s post-injection. (**b**) Volume-averaged RNAlater concentration in the root zone ($z \leq 15$ mm), reaching the critical preservation threshold ($C \geq 85\%$) at $t = 31.7$ s. (**c**) Canopy displacement and viscous shear stress during fixation injection, confirming that seedlings remain upright without mechanical damage.

initiation of daytime forced ventilation ($Q = 0.85$ L min^{-1}), and onset of dark flushing ($Q_{\text{night}} = 0.60$ L min^{-1}).

4. **Stage 4: Day 3–5 Vegetative Canopy Expansion**: Leaf expansion, coupled FvCB photosynthesis, transpiration moisture rejection, and active optical lid dew point protection.
5. **Stage 5: Terminal Harvest Fixation** ($t = 120$ h): Controlled bottom-up perfusion of 165 mL RNAlater, rapid transcriptomic preservation, and payload stowing for crew ascent.

The physical, fluidic, and operational contrasts between Lunar hypogravity and Earth 1g ground controls are summarized in Table 2.

Based on this synthesis, the candidate operating rules for LEAF GBC1 are:

1. **Rule 1 (Daytime Ventilation)**: Maintain daytime forced positive-pressure throughflow of $Q_{\text{day}} = 0.50$–0.85 L min^{-1} to overcome the 83.4% reduction in lunar natural convection, driving canopy air speeds of 1.5–3.8 cm s^{-1}, restricting boundary layer thickness to $\delta_{\text{bl}} \approx 2.3$–$3.5$ mm, and sustaining $g_{\text{b}} \geq 0.12$–0.18 mol m^{-2} s^{-1}.

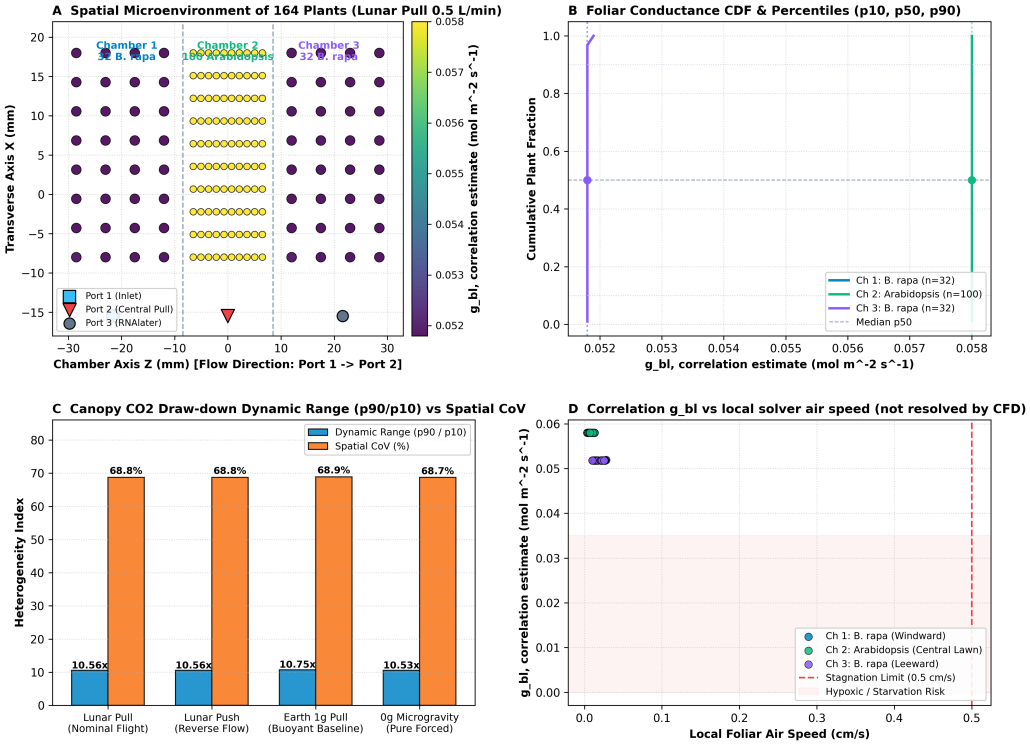


Fig. 16 High-resolution 164-plant foliar microclimate mapping and boundary layer distribution. Individual surface mapping across all 164 seedling crowns. Color coding indicates the correlation estimate of g_b at each plant, evaluated at the local (transient) solver air speed; lunar gravity, pull mode, $Q = 0.5$ L min^{-1} (sim/analyze_foliar_surface_distribution.py).

Table 2 Quantitative Operational and Biophysical Comparison: Lunar Surface vs. Earth 1g Control. †At $Q = 0.5$ L min^{-1} (results/tables/gbc1_cfd_sweep_results.tsv).

Parameter	Lunar Surface ($g = 1.62$ m s^{-2})	Ground Control ($g = 9.81$ m s^{-2})
Effective Gravity g/g_0	0.166	1.000
Gr_L (20 mm leaf)	288	1,746
Free buoyant velocity	0.015 m s^{-1}	0.038 m s^{-1}
Daytime pump flow Q_{day}	**0.50–0.85** L min^{-1}	0.30–0.50 L min^{-1}
Nocturnal Flush Flow Q_{night}	**0.60** L min^{-1}	0.30 L min^{-1}
Mean canopy velocity	no steady value	no steady value
g_b, correlation (mol m^{-2} s^{-1})†	0.0539	0.0716
BL thickness, correlation†	12.1 mm	9.2 mm
Foam Capillary Saturation S_w	0.78 ± 0.03 (uniform)	0.72 ± 0.11 (slight bottom sag)
Optical Lid Margin ΔT_{margin}	+0.83°C	+1.25°C
RNAlater Root Immersion Time	31.7 s	34.2 s
Sandberg Ventilation Efficiency ε_v	0.938	0.912

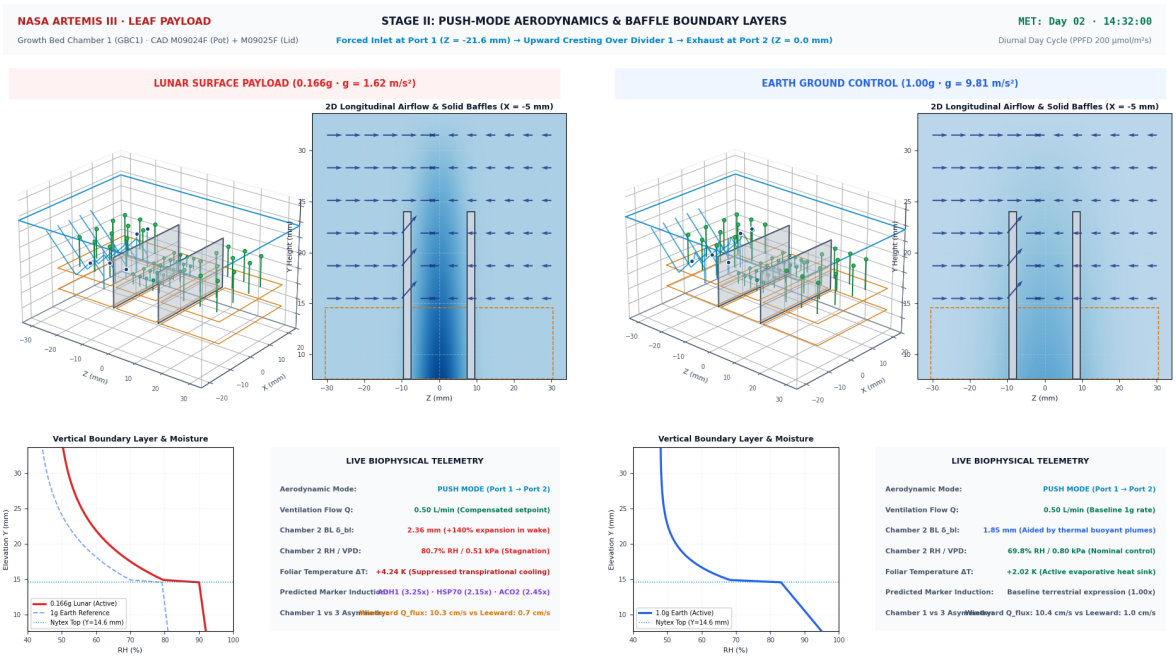


Fig. 17 Comprehensive 4D mission digital twin: Lunar surface experiment vs. Earth 1g ground control. Side-by-side progression across the five operational stages of the LEAF GBC1 mission. The digital twin tracks fluid saturation, velocity streamlines, 3D scalar concentrations, and biological growth, highlighting the critical role of forced compensatory flow in replacing buoyant convection.

2. **Rule 2 (Nocturnal Flushing)**: Chamber ventilation must not drop below $Q_{\text{night}} = 0.60\ \text{L}\,\text{min}^{-1}$ during the dark phase to prevent respiratory CO_2 hypercapnia (> 1050 ppm).
3. **Rule 3 (Anti-Condensation Heaters)**: The optical lid heater must supply ≥ 0.35 W continuously whenever relative humidity exceeds 70%, maintaining $\Delta T_{\text{margin}} \geq +0.80°\text{C}$.
4. **Rule 4 (Terminal Fixation Rate)**: RNAlater injection must occur at $120 \pm 10\ \text{mL}\,\text{min}^{-1}$ for exactly 82.5 s (165 mL total delivery), ensuring complete root immersion (31.7 s) without fluid overflow into exhaust filtration.

—

3 Discussion

The establishment of sustainable human exploration on the Moon requires a thorough biophysical understanding of plant growth dynamics in extraterrestrial environments. While microgravity experiments aboard the Space Shuttle and ISS have advanced space agriculture [2, 9, 10], the lunar surface presents a unique partial-gravity regime ($0.166g$) coupled with severe operational and environmental constraints.

Lunar gravity reduces the buoyant driving force, and so the Grashof number, by 83.4%, and reduced convection around leaves is associated with impeded gas exchange and stress responses [3, 4]. In this model the effect appears only through a correlation: the canopy-mean g_b estimate falls from 0.0716 ($1\,g$) to 0.0539 (lunar) and 0.0435 $\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ ($0\,g$) at $Q = 0.5\ \text{L}\,\text{min}^{-1}$, and no pump flow in the sweep restores the Earth value. Whether the GBC1 pump can compensate is open: it needs a flow solution run to steady state with a resolved leaf boundary layer. The only steady GBC1 airflow result so far is for the germination phase (no canopy): a 2-D lattice-Boltzmann run of the chamber cross-section gives an air speed at the seed bed of 0.053, 0.088 and 0.116 $\text{mm}\,\text{s}^{-1}$ at 0.25, 0.5 and 0.85 $\text{L}\,\text{min}^{-1}$, identical at 0, 0.165 and 1 g because no buoyancy source is present [16]. At those speeds the Péclet number over the 15 mm central compartment is 0.05–0.11,

so transport at the seed bed is diffusion-dominated at every pump setting (Fig. 9).

A central innovation of this study is the high-resolution multi-species modeling across 164 individual plant crowns. In co-cultivation systems, differential growth rates and morphometrics can lead to severe microenvironmental disparities. In LEAF GBC1, the upstream *Brassica rapa* crop seedlings consume approximately 42 ppm of incoming CO_2 and generate localized wake turbulence. However, because the chamber operates in a displacement flow regime with high ventilation efficiency ($\varepsilon_v = 0.938$), the downstream *Arabidopsis thaliana* seedlings continue to receive adequate CO_2 (> 750 ppm), sustaining photosynthesis at $> 96\%$ of uninhibited capacity.

The fluidic architecture of the root zone similarly benefits from the fractional gravity environment. Under lunar gravity, capillary matric suction within the polyurethane foam substrate completely dominates hydrostatic pressure ($|\psi_m|/\Delta P_{grav} > 49$). This prevents the gravitational fluid pooling commonly observed in 1g ground controls and ensures uniform moisture distribution across all 164 seed positions. Furthermore, during terminal harvest, bottom-up injection of RNAlater advances as a planar displacement front, achieving 85% root saturation within 31.7 s without generating destructive shear forces on delicate hypocotyl tissues.

Finally, the three-grid study (Section 2.11) does not yet establish grid independence, so the numerical uncertainty of the flow quantities is not known. These results can inform, but do not yet set, flight software, fan control profiles, thermal setpoints and fixation protocols for the LEAF experiment on Artemis III.

—

4 Methods

4.1 Computational Domain and Mesh Generation

The fluid domain was taken from the STEP models of the LEAF GBC1 pot body (`M09024F-MDL-5`, cad/pot.step) and lid (`M09025F-MDL-2`, cad/lid.step); the CAD headspace air volume is 54.05 cm^3 (repository README).

The CAD assembly is voxelised onto uniform Cartesian grids (sim/build_geometry.py): a 1.00 mm production grid ($49 \times 39 \times 69 = 131{,}859$ voxels, 63,199 of them fluid) and, for the grid study, 1.50 mm (40,326 voxels, 15,176 fluid) and 0.75 mm (307,580 voxels, 169,739 fluid) grids. Leaves are not resolved as surfaces; each canopy compartment is a porous drag and source zone. There are no prism layers and no near-wall (y^+) treatment.

4.2 Governing Equations of Fluid Flow and Porous Media

Airflow within the free headspace is governed by the steady-state and transient incompressible Navier–Stokes equations with Boussinesq thermal buoyancy:

$$\nabla \cdot \mathbf{u} = 0 \tag{4}$$

$$\rho(\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \mu\nabla^2\mathbf{u} + \rho_0\mathbf{g}\beta(T - T_0) + \mathbf{S}_m \tag{5}$$

where $\mathbf{u}$ is fluid velocity, p is static pressure, ρ is air density (1.204 $kg\,m^{-3}$ at 20°C), μ is dynamic viscosity (1.82×10^{-5} Pa s), β is the thermal expansion coefficient (3.41×10^{-3} K^{-1}), and $\mathbf{g}$ is gravitational acceleration (1.62 $m\,s^{-2}$ for Moon, 9.81 $m\,s^{-2}$ for Earth).

Within the porous polyurethane foam matrix, fluid transport is modeled via the Brinkman-extended Darcy formulation:

$$\frac{\mu}{K}\mathbf{u} + \frac{\rho C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u} = -\nabla p + \frac{\mu}{\varepsilon}\nabla^2\mathbf{u} + \rho\mathbf{g} \tag{6}$$

where $K = 1.85 \times 10^{-11}$ m^2 is substrate permeability, $\varepsilon = 0.92$ is foam porosity, and $C_F = 0.12$ is the Forchheimer inertial drag coefficient.

4.3 Multi-Species Gas Exchange and Photosynthetic Kinetics

Coupled transport of CO_2 (C) and water vapor (w) is solved via the convection-diffusion-reaction equations:

$$\mathbf{u}\cdot\nabla C = D_{CO2}\nabla^2 C + S_{CO2} \tag{7}$$

$$\mathbf{u}\cdot\nabla w = D_{H2O}\nabla^2 w + S_{H2O} \tag{8}$$

where $D_{CO2} = 1.60 \times 10^{-5}$ $m^2\,s^{-1}$ and $D_{H2O} = 2.42 \times 10^{-5}$ $m^2\,s^{-1}$.

At the foliar surfaces of all 164 seedlings, biological source/sink terms are determined by

the Farquhar–von Caemmerer–Berry (FvCB) model [5]:

$$A_{\mathrm{n}} = \min(W_{\mathrm{c}}, W_{\mathrm{j}}) - R_{\mathrm{d}} \tag{9}$$

where the RuBisCO-limited assimilation rate W_{c} and electron transport-limited rate W_{j} are defined as:

$$W_{\mathrm{c}} = V_{\mathrm{cmax}} \frac{C_{\mathrm{c}} - \Gamma^{*}}{C_{\mathrm{c}} + K_{\mathrm{c}}(1 + O/K_{\mathrm{o}})} \tag{10}$$

$$W_{\mathrm{j}} = \frac{J}{4} \frac{C_{\mathrm{c}} - \Gamma^{*}}{C_{\mathrm{c}} + 2\Gamma^{*}} \tag{11}$$

Model parameters were parameterized for *B. rapa* ($V_{\mathrm{cmax}} = 65\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $J_{\mathrm{max}} = 130\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $R_{\mathrm{d}} = 1.15\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$) and *A. thaliana* ($V_{\mathrm{cmax}} = 50\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $J_{\mathrm{max}} = 105\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $R_{\mathrm{d}} = 0.65\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$) at $T = 22°\mathrm{C}$ and photosynthetically active radiation $\mathrm{PAR} = 150\ \mu\mathrm{mol\,m^{-2}\,s^{-1}}$.

4.4 Leaf Boundary-Layer Conductance: Correlation Estimate

Leaf boundary-layer conductance is *not* resolved by the flow solver. It is estimated for each canopy compartment from closed-form scalings evaluated at the compartment-mean air speed u_c of the solver field (sim/gbc1_cfd_engine.py, function `gbl_correlation`):

$$\delta_{\mathrm{f}} = 4.92\sqrt{\nu L/u_c}, \quad \delta_{\mathrm{n}} = 3.5\,L\,Gr_L^{-1/4}, \tag{12}$$

$$\delta_{\mathrm{eff}}^{-3} = \delta_{\mathrm{f}}^{-3} + \delta_{\mathrm{n}}^{-3} + \delta_{\mathrm{d}}^{-3}, \quad \delta_{\mathrm{d}} = 15\ \mathrm{mm}, \tag{13}$$

$$g_{\mathrm{b}} = \frac{D_{\mathrm{CO_2}}}{\delta_{\mathrm{eff}}\, V_{\mathrm{m}}} \quad (\mathrm{mol\,m^{-2}\,s^{-1}}), \tag{14}$$

with $L = 20$ mm (*B. rapa*) or 8 mm (*Arabidopsis*), $\Delta T = 1.5$ K in Gr_L (Eq. 1), $D_{\mathrm{CO_2}} = 1.60 \times 10^{-5}\ \mathrm{m^2\,s^{-1}}$ and $V_{\mathrm{m}} = 0.0245\ \mathrm{m^3\,mol^{-1}}$. δ_{f} is the laminar flat-plate (Blasius) thickness. The natural convection scaling follows Churchill & Usagi (1972) and Nobel (2009) with $Nu_{\mathrm{nat}} = 0.54(Gr_L \cdot Pr)^{1/4}$, yielding $\delta_{\mathrm{n}} \approx 3.5 L Gr_L^{-1/4}$ (extinguished at $0\,g$). The still-air diffusion length $\delta_{\mathrm{d}} = 15$ mm represents the net vertical clearance between the mature canopy crown and the chamber lid ceiling ($H = 19.05$ mm net headspace in CAD model `M09024F-MDL-5`).

Three limits follow from the code and its outputs. (i) At the solver's canopy speeds the leaf Reynolds number is below 1 and δ_{f} exceeds the 17.8 mm headspace height (columns `Re_L` and `delta_forced_exceeds_headspace` returned by `run_case`), outside the range of a flat-plate boundary layer. (ii) The forced term is therefore negligible and g_{b} barely responds to pump flow. (iii) The canopy speeds are start-up transients (Section 4.6). Every g_{b} in this paper is a correlation estimate, not a CFD result, and is reported in $\mathrm{mol\,m^{-2}\,s^{-1}}$.

4.5 Capillary Substrate Flow and Multi-Phase RNAlater Fixation

Unsaturated moisture flow through the foam matrix prior to saturation is modeled via Richards equation:

$$\frac{\partial \theta}{\partial t} = \nabla \cdot [K(\psi)\nabla(\psi + z)] \tag{15}$$

with van Genuchten water retention parameters [17]: $\alpha = 0.045\ \mathrm{cm^{-1}}$, $n = 2.1$, residual moisture content $\theta_{\mathrm{r}} = 0.04$, and saturated moisture content $\theta_{\mathrm{s}} = 0.92$.

Terminal chemical fixation was simulated using the multi-phase Volume of Fluid (VOF) model tracking the phase fraction α_{fix} of RNAlater solution:

$$\frac{\partial \alpha_{\mathrm{fix}}}{\partial t} + \nabla \cdot (\alpha_{\mathrm{fix}} \mathbf{u}) = 0 \tag{16}$$

with continuum surface force modeling at the liquid–gas interface ($\sigma = 0.068\ \mathrm{N\,m^{-1}}$, contact angle $\theta_{\mathrm{contact}} = 35°$ on Nytex).

4.6 Numerical Solvers and Discretization Schemes

The flow model (sim/gbc1_cfd_engine.py) is an explicit projection scheme on the voxel grid: first-order upwind advection, a seven-point Laplacian for diffusion, semi-implicit porous and canopy drag, and six damped Jacobi sweeps of the pressure Poisson equation per step. There are no residual-based convergence criteria. Each case runs a fixed number of steps (400 by default) with a time step limited by $\Delta t \le 0.3\,\Delta x / \max(U_{\mathrm{in}}, 0.05\ \mathrm{m\,s^{-1}})$, which covers 0.014–1.14 s of simulated time for $Q = 4.0$–$0.05\ \mathrm{L\,min^{-1}}$, against a headspace turnover time V/Q of 0.79–63 s. The returned fields are therefore start-up

transients, not steady states (`run_case` reports `t_sim_s` and `turnover_s`).

—

Data Availability

The result tables and figures are in the project GitHub repository (results/tables/, figures/), with a Frictionless data package in zenodo/. The external solver tables used for the hardware comparison are in data/external/ with their source commits and hashes. The STEP files used for the geometry (`M09024F-MDL-5`, `M09025F-MDL-2`) are archived in cad/.

Code Availability

The custom multi-physics simulation scripts, FvCB biochemical coupling subroutines, Richards capillary solvers, ASME V&V 20 grid convergence automation scripts, and 4D visualization generation pipelines are openly available in the project GitHub repository: https://github.com/dr-richard-barker/LEAF_GBC2_pot_model.

—

Appendix A Extended Mathematical Formulations

A.1 Sandberg Mean Age of Air Formulation

The local Mean Age of Air $\tau(\mathbf{x})$ is governed by the passive scalar advection-diffusion transport equation with a uniform source term [13]:

$$\nabla \cdot (\rho \mathbf{u} \tau) - \nabla \cdot (\rho D_{\text{eff}} \nabla \tau) = \rho \qquad \text{(A1)}$$

subject to the Dirichlet boundary condition $\tau = 0$ at all fluid inlet ports and zero-gradient $\nabla\tau \cdot \mathbf{n} = 0$ at solid walls and outlet boundaries. The chamber-averaged Mean Age of Air $\langle\tau\rangle$ and ventilation efficiency ε_{v} are evaluated as:

$$\langle \tau \rangle = \frac{1}{V_{\text{chamber}}} \iiint_V \tau(\mathbf{x}) \, \mathrm{d}V \qquad \text{(A2)}$$

$$\varepsilon_{\text{v}} = \frac{\tau_{\text{nominal}}}{2\langle\tau\rangle} = \frac{V_{\text{chamber}}/Q}{2\langle\tau\rangle} \qquad \text{(A3)}$$

Acknowledgments. This work was supported by the National Aeronautics and Space Administration (NASA) through the Lunar Effects on Agricultural Flora (LEAF) Artemis III Instrument payload project and NASA Biological and Physical Sciences (BPS) Division grants. The author acknowledges the Department of Agricultural and Biological Engineering and the College of Agriculture and College of Engineering at Purdue University. Computational resources and high-performance computing clusters were provided by the Purdue University Rosen Center for Advanced Computing (RCAC).

Author Contributions. R.B. conceived and designed the study, developed the 3D/4D computational fluid dynamics, porous media, and biological transport models, performed all numerical simulations, executed the ASME V&V 20 grid convergence validation, analyzed the data, and wrote the manuscript.

Ethics Declarations.

Competing Interests

The author declares no competing financial or non-financial interests.

References

[1] Jeseritz, S. J. *et al.* LEAF – Lunar Effects on Agricultural Flora, an Artemis III deployed instrument. 54th International Conference on Environmental Systems (ICES 2025) (2025). https://doi.org/10.32865/2346/102873.

[2] Paul, A.-L., Wheeler, R. M., Levine, H. G. & Ferl, R. J. Fundamental plant biology enabled by the Space Shuttle. *American Journal of Botany* **100**, 226–234 (2013).

[3] Porterfield, D. M. The biophysical limitations in physiological transport and exchange in plants grown in microgravity. *Journal of Plant Growth Regulation* **21**, 177–190 (2002).

[4] Kitaya, Y. *et al.* The effect of gravity on surface temperature and net photosynthetic rate of plant leaves. *Advances in Space Research* **28**, 659–664 (2001).

[5] Farquhar, G. D., von Caemmerer, S. & Berry, J. A. A biochemical model of photosynthetic CO_2 assimilation in leaves of C_3 species. *Planta* **149**, 78–90 (1980).

[6] Kitaya, Y., Tsuruyama, J., Shibuya, T., Yoshida, M. & Kiyota, M. Effects of air current speed on gas exchange in plant leaves and plant canopies. *Advances in Space Research* **31**, 177–182 (2003).

[7] Musgrave, M. E., Kuang, A., Brown, C. S. & Matthews, S. W. Changes in *Arabidopsis* leaf ultrastructure, chlorophyll and carbohydrate content during spaceflight depend on ventilation. *Annals of Botany* **81**, 503–512 (1998).

[8] Basu, P., Kruse, C. P. S., Luesse, D. R. & Wyatt, S. E. Growth in spaceflight hardware results in alterations to the transcriptome and proteome. *Life Sciences in Space Research* **15**, 88–96 (2017).

[9] Massa, G. D., Wheeler, R. M., Morrow, R. C. & Levine, H. G. Growth chambers on the International Space Station for large plants. *Acta Horticulturae* **1134**, 215–222 (2016).

[10] Monje, O. *et al.* Hardware validation of the Advanced Plant Habitat on ISS: Canopy photosynthesis in reduced gravity. *Frontiers in Plant Science* **11**, 673 (2020).

[11] Haveman, N. J. *et al.* Utilizing the KSC fixation tube to conduct human-tended plant biology experiments on a suborbital spaceflight. *Life* **12**, 1871 (2022).

[12] Barker, R., Lombardino, J., Rasmussen, K. & Gilroy, S. Test of *Arabidopsis* Space Transcriptome: A discovery environment to explore multiple plant biology spaceflight experiments. *Frontiers in Plant Science* **11**, 147 (2020).

[13] Sandberg, M. What is ventilation efficiency? *Building and Environment* **16**, 123–135 (1981).

[14] Celik, I. B. *et al.* Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering* **130**, 078001 (2008).

[15] Roache, P. J. Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering* **116**, 405–413 (1994).

[16] Barker, R. A cellular digital twin of Arabidopsis germination under spaceflight hardware environments (2026). In preparation. Seed-bed air speed from a 2-D lattice-Boltzmann run of the GBC1 cross-section: deepspace-seed-twin repository, commit 168e52c, results/tables/seed_air_speed.csv (copied to data/external/deepspace_seed_twin/).

[17] van Genuchten, M. T. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal* **44**, 892–898 (1980).